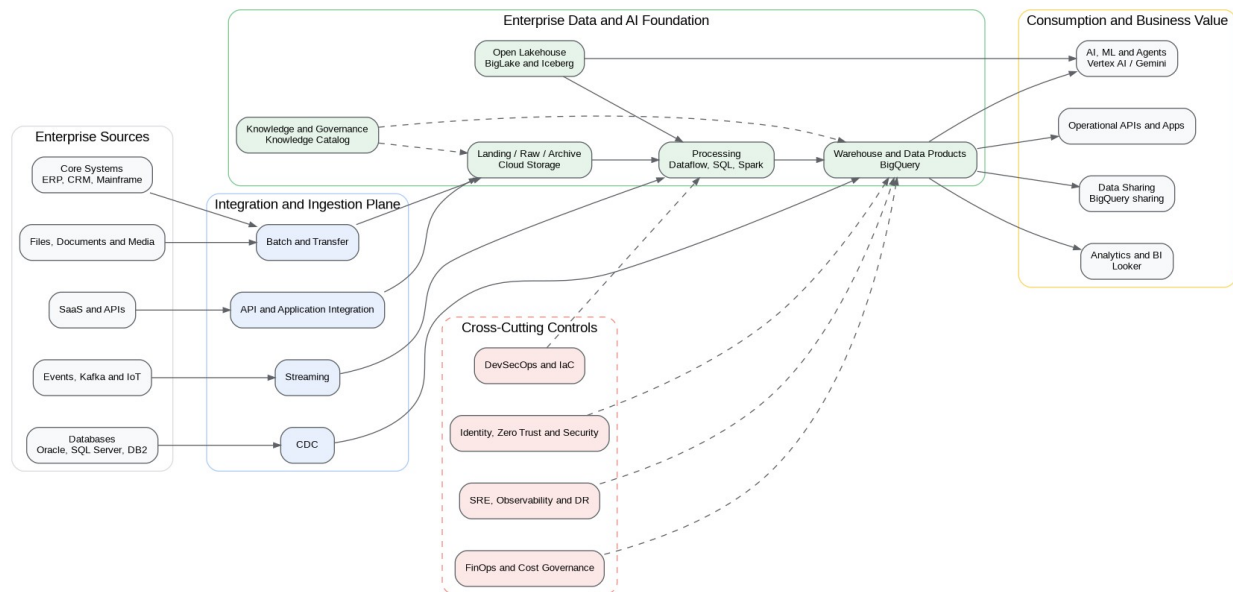


ENTERPRISE GOOGLE CLOUD DATA PLATFORM BLUEPRINT

VOLUME 01

Enterprise Architecture Overview



Fortune 100 Enterprise Data and AI Platform - Conceptual View

Implementation-ready reference architecture

Fortune 100 scale | Petabyte data estate | Global regions | Regulated workloads

Version 1.0 | July 2026

Document Control

This volume establishes the business, architectural and technology foundation for the complete Enterprise Google Cloud Data Platform Blueprint Series. It is a reusable reference architecture, not a substitute for organization-specific discovery and design authority.

Section outcomes

establish a common enterprise baseline; define measurable architecture requirements; record major technology decisions; provide a staged adoption roadmap.

Field	Value
Document title	Enterprise Google Cloud Data Platform Blueprint - Volume 01: Enterprise Architecture Overview
Document type	Enterprise reference architecture and transformation guide
Version	1.0
Status	Reference baseline for architecture review
Audience	CIO, CTO, CDO, CISO, enterprise architects, platform leaders, data leaders, engineering and FinOps
Scope	Business drivers through target reference architecture, technology decisions and roadmap
Out of scope	Detailed deployment code, workload-specific low-level design and production change records
Assurance cadence	Quarterly architecture review and event-driven review after material platform changes
Source prompt	Enterprise Google Cloud Data Platform Blueprint Series master prompt supplied by the user

Revision History

Version	Date	Authoring body	Change
0.1	July 2026	Enterprise Architecture Team	Initial working draft
0.9	July 2026	Architecture Review Board	Quality and consistency review
1.0	July 2026	Enterprise Data and AI Council	Reference baseline issued

How to Use This Volume

This guide is written for a large global enterprise that must reconcile speed, regulatory control, open interoperability, regional data sovereignty and cost accountability. It provides a preferred architecture together with explicit exceptions, decision criteria and acceptance evidence.

Architecture teams should use the requirements and principles as review criteria. Platform engineering should use the service mappings and roadmap as a planning baseline. Domain teams should use the capability model to identify which responsibilities are supplied centrally and which remain with data-product teams.

All product capabilities, locations, quotas, pricing constructs and regulatory mappings must be validated against current authoritative documentation before production implementation. The references section identifies the primary Google Cloud sources used for this baseline.

Contents

The document is organized as a decision sequence: why the enterprise needs the platform, what the platform must achieve, the principles that constrain design, the target architecture, the decisions that shape implementation and the roadmap that converts architecture into delivery.

Section outcomes

executive alignment; architecture traceability; technology governance; delivery sequencing.

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Document scale

This volume is intentionally detailed and is designed to exceed seventy rendered pages. Detailed landing-zone implementation, pipeline code, security control mapping and industry-specific designs are developed in subsequent volumes.

1. Executive Summary

The enterprise data platform is positioned as a governed business capability rather than a collection of cloud services. Its objective is to make trusted data and AI context available at the speed required by business operations while maintaining defensible security, privacy, resilience and financial controls.

Section outcomes

align executives on the target state; connect investment to measurable outcomes; define the recommended architecture stance; highlight decisions and risks requiring sponsorship.

1.1 Strategic Imperative

The enterprise is accumulating data faster than existing operating models can govern, integrate and convert into business outcomes. Fragmented warehouses, local data lakes, uncoordinated streaming platforms and duplicated analytics create inconsistent metrics, rising operating cost and material regulatory exposure. The proposed platform creates a common foundation that separates enterprise controls from domain delivery, enabling business units to move faster without rebuilding security, ingestion and governance capabilities for every initiative.

The transformation should be sponsored as a multi-year product program with stable funding, executive data ownership and measurable adoption targets. A project-only approach will create a technically capable platform that lacks the product management, domain accountability and operational ownership required for sustainable use.

1.2 Recommended Architecture Position

Adopt BigQuery as the primary governed analytical platform, Cloud Storage and BigLake for open-format lakehouse patterns, Pub/Sub and Dataflow for event-driven processing, Datastream for supported CDC, Dataform for BigQuery ELT, Managed Service for Apache Airflow for complex orchestration, Knowledge Catalog for data and AI context, Looker for governed semantics and Vertex AI for ML and generative AI.

The architecture is serverless-first and managed-service-first. It does not prohibit Spark, Kafka, Kubernetes or third-party tools; it requires each exception to demonstrate a capability, portability or migration need that outweighs the operational and security burden of a self-managed service.

1.3 Operating Model

Use a centrally governed, federated delivery model. A central platform product team owns landing-zone patterns, shared ingestion, reusable pipeline templates, catalog integration, security controls, reliability standards and FinOps. Domain teams own data meaning, source relationships, quality, product interfaces, access approval and service levels.

This model prevents the central team becoming a data-delivery bottleneck while avoiding the uncontrolled duplication that occurs when every business unit builds its own cloud platform. Governance is implemented computationally through templates, policy checks and metadata requirements rather than through manual review alone.

1.4 Security and Compliance Position

Establish identity, network, workload and data perimeters. Use group-based IAM, short-lived workload credentials, organization policies, Shared VPC, Private Service Connect, VPC Service Controls where supported, centralized KMS and secrets, comprehensive audit logging and automated classification.

Controls must be proportional to data classification. Restricted data requires stronger key, perimeter, masking, approval and monitoring controls than internal operational data. The architecture avoids a one-size-fits-all control model that either leaves regulated data exposed or makes low-risk data impractical to use.

1.5 Resilience Position

Design critical ingestion and product interfaces for multi-zone operation and define cross-region recovery for Tier 0 and Tier 1 data products. Use replayable event logs, immutable raw storage, declarative infrastructure, source-controlled transformation logic and tested recovery runbooks.

Managed-service availability does not eliminate enterprise recovery responsibility. Recovery objectives must be defined by data product and validated with exercises that test data, metadata, configuration, identities, network paths and operational communications.

1.6 Financial Position

Implement cost ownership at the domain and data-product level. Use budgets, billing export, standardized labels, BigQuery workload isolation, capacity planning, storage lifecycle policies, logging controls and unit-cost metrics.

FinOps is embedded in architecture decisions. Teams must understand not only monthly service spend but also cost per terabyte ingested, per data product, per thousand events, per analytical workload and per AI interaction.

1.7 AI Readiness

Treat governed metadata, semantic definitions, lineage, quality and access policies as the foundation for AI. AI agents and RAG applications should retrieve from certified sources, apply identity-aware access, preserve citations and log model, prompt and retrieval behavior.

The enterprise should not create a separate ungoverned AI data estate. Vertex AI and Gemini capabilities consume the same certified products, classifications and regional controls used by analytics and operational applications.

1.8 Delivery Recommendation

Deliver the platform in waves: mobilize, establish the foundation, release a minimum viable platform, prove lighthouse domains, industrialize reusable patterns, migrate at scale and optimize with AI-assisted operations.

The first production release should include one representative batch pipeline, one CDC pipeline, one streaming use case, one restricted dataset, one certified data product, one BI product and one AI-readiness use case. This scope proves the cross-cutting platform rather than only a single technology component.

1.9 Executive Outcome Scorecard

Outcome	Baseline challenge	36-month target	Executive owner
Trusted data availability	Weeks to months to locate and approve data	80% of strategic datasets cataloged; 60% certified	Chief Data Officer
Delivery speed	Bespoke platform setup for each program	Standard domain environment in less than 10 business days	CTO / Platform Engineering
Real-time capability	Limited event integration and duplicated Kafka stacks	Tier 0 events available to approved consumers within agreed seconds-level SLO	Chief Digital Officer
Security assurance	Inconsistent project, key and identity controls	100% production workloads deployed through approved guardrails	CISO
Resilience	Recovery plans focus on infrastructure, not data products	Tier 0 and Tier 1 recovery exercises pass twice yearly	COO / SRE
Cost transparency	Shared cloud bill with limited ownership	95% of platform cost allocated to domain, product and environment	CFO / FinOps
AI readiness	Unstructured experiments use inconsistent context	Certified AI-ready products and governed retrieval patterns	Chief AI Officer

1.10 Executive Decisions Required

19. Approve the central platform product model and recurring funding mechanism.
20. Assign accountable data owners for each priority business domain.
21. Approve the regional data-residency and paired-recovery strategy.
22. Confirm BigQuery as the default analytical platform and approve an exception process.
23. Approve the classification-driven security model and selective CMEK policy.
24. Approve showback in year one and chargeback for mature domains from year two.
25. Nominate lighthouse domains and commit business resources to data-product ownership.

26. Approve target legacy platform retirement dates and migration incentives.

2. Business Drivers

The target platform is justified by a combination of growth, resilience, regulatory, operating-model and artificial-intelligence imperatives. The drivers below establish the business outcomes against which architecture choices and investment decisions will be measured.

Section outcomes

A traceable case for change; Measurable outcomes; Architecture implications; Executive ownership.

2.1 BD-01 — Accelerate decision velocity

Business leaders require trusted operational and strategic information in minutes rather than through weekly or monthly reporting cycles. **Architecture response:** Reduce the elapsed time from source event to governed decision-ready data; standardize reusable metrics; support event-driven actions where latency has demonstrable business value.

Dimension	Definition
Business outcome	Accelerate decision velocity
Architecture implication	Reduce the elapsed time from source event to governed decision-ready data; standardize reusable metrics; support event-driven actions where latency has demonstrable business value.
Measures	Median time to onboard a priority source; percentage of certified products meeting freshness SLO; time required to introduce a governed KPI.
Executive accountability	Chief Data Officer and business-unit executives
Decision consequence	Initiatives that do not map to this driver require an explicit exception or a separate approved business case.

The driver is treated as an investment constraint rather than a marketing statement. Architecture reviews must demonstrate which measurable outcome will improve, the baseline used, the target date, and the accountable owner. Benefits are tracked through the platform product backlog and quarterly value reviews.

2.2 BD-02 — Create a reusable enterprise data foundation

Business units currently build overlapping ingestion, transformation and governance capabilities, creating duplication and inconsistent control. **Architecture response:** Provide centrally engineered platform capabilities with domain-configurable templates for ingestion, quality, metadata, security, observability and delivery.

Dimension	Definition
Business outcome	Create a reusable enterprise data foundation
Architecture implication	Provide centrally engineered platform capabilities with domain-configurable templates for ingestion, quality, metadata, security, observability and delivery.
Measures	Reuse rate of platform modules; reduction in duplicate tools; percentage of pipelines using approved templates.
Executive accountability	CIO, CDO and Platform Product Owner
Decision consequence	Initiatives that do not map to this driver require an explicit exception or a separate approved business case.

The driver is treated as an investment constraint rather than a marketing statement. Architecture reviews must demonstrate which measurable outcome will improve, the baseline used, the target date, and the accountable owner. Benefits are tracked through the platform product backlog and quarterly value reviews.

2.3 BD-03 — Modernize legacy data warehouses and Hadoop estates

Legacy warehouses and distributed data platforms carry high license, infrastructure and specialist-support costs and constrain elastic scale. **Architecture response:** Adopt BigQuery, Cloud Storage and BigLake as the default analytical foundation while using controlled coexistence and migration waves to protect business continuity.

Dimension	Definition
Business outcome	Modernize legacy data warehouses and Hadoop estates
Architecture implication	Adopt BigQuery, Cloud Storage and BigLake as the default analytical foundation while using controlled coexistence and migration waves to protect business continuity.
Measures	Workloads retired; legacy run-cost avoided; migrated query performance; reconciliation pass rate.
Executive accountability	CIO, Application Modernization Lead and Finance
Decision consequence	Initiatives that do not map to this driver require an explicit exception or a separate approved business case.

The driver is treated as an investment constraint rather than a marketing statement. Architecture reviews must demonstrate which measurable outcome will improve, the baseline used, the target date, and the accountable owner. Benefits are tracked through the platform product backlog and quarterly value reviews.

2.4 BD-04 — Enable real-time and event-driven operations

Fraud, customer experience, supply-chain and operational use cases lose value when data is available only after batch windows. **Architecture response:** Use Pub/Sub and Dataflow for justified streaming patterns, with CDC through Datastream and explicit replay, idempotency and dead-letter controls.

Dimension	Definition
Business outcome	Enable real-time and event-driven operations
Architecture implication	Use Pub/Sub and Dataflow for justified streaming patterns, with CDC through Datastream and explicit replay, idempotency and dead-letter controls.
Measures	Event-to-insight latency; replay success rate; streaming SLO attainment; percentage of streams with defined owners.
Executive accountability	Business Operations and Digital Product Leaders
Decision consequence	Initiatives that do not map to this driver require an explicit exception or a separate approved business case.

The driver is treated as an investment constraint rather than a marketing statement. Architecture reviews must demonstrate which measurable outcome will improve, the baseline used, the target date, and the accountable owner. Benefits are tracked through the platform product backlog and quarterly value reviews.

2.5 BD-05 — Strengthen regulatory compliance and privacy

The enterprise processes sensitive personal, financial, health and employee data across jurisdictions with different retention, access and residency obligations. **Architecture response:** Implement policy-driven classification, fine-grained access, VPC Service Controls, auditability, selective CMEK, retention controls and privacy workflows.

Dimension	Definition
Business outcome	Strengthen regulatory compliance and privacy
Architecture implication	Implement policy-driven classification, fine-grained access, VPC Service Controls, auditability, selective CMEK, retention controls and privacy workflows.
Measures	Control coverage; access-review completion; privacy-request cycle time; unresolved audit findings.
Executive accountability	Chief Information Security Officer, Privacy Officer and General Counsel
Decision consequence	Initiatives that do not map to this driver require an explicit exception or a separate approved business case.

The driver is treated as an investment constraint rather than a marketing statement. Architecture reviews must demonstrate which measurable outcome will improve, the baseline used, the target date, and the accountable owner. Benefits are tracked through the platform product backlog and quarterly value reviews.

2.6 BD-06 — Support global sovereignty and regional autonomy

Regional regulations and business-continuity requirements prevent a single global data location from satisfying every workload. **Architecture response:** Use regional data-residency zones, explicit replication policies and globally consistent control standards while keeping data products discoverable across the enterprise.

Dimension	Definition
Business outcome	Support global sovereignty and regional autonomy
Architecture implication	Use regional data-residency zones, explicit replication policies and globally consistent control standards while keeping data products discoverable across the enterprise.
Measures	Residency-policy compliance; approved cross-region transfers; regional recovery-test results.
Executive accountability	Regional CIOs, Security and Data Governance
Decision consequence	Initiatives that do not map to this driver require an explicit exception or a separate approved business case.

The driver is treated as an investment constraint rather than a marketing statement. Architecture reviews must demonstrate which measurable outcome will improve, the baseline used, the target date, and the accountable owner. Benefits are tracked through the platform product backlog and quarterly value reviews.

2.7 BD-07 — Operate data as a product

Datasets are often delivered without ownership, documentation, quality targets or support commitments, reducing trust and adoption. **Architecture response:** Assign accountable domain owners, publish contracts and SLOs, certify products through Knowledge Catalog, and expose approved products through governed interfaces.

Dimension	Definition
Business outcome	Operate data as a product
Architecture implication	Assign accountable domain owners, publish contracts and SLOs, certify products through Knowledge Catalog, and expose approved products through governed interfaces.

Dimension	Definition
Measures	Certified-product count; product adoption; quality SLO attainment; consumer satisfaction.
Executive accountability	Domain Executives and Data Product Owners
Decision consequence	Initiatives that do not map to this driver require an explicit exception or a separate approved business case.

The driver is treated as an investment constraint rather than a marketing statement. Architecture reviews must demonstrate which measurable outcome will improve, the baseline used, the target date, and the accountable owner. Benefits are tracked through the platform product backlog and quarterly value reviews.

2.8 BD-08 — Create an AI-ready and agent-ready foundation

AI initiatives are slowed by fragmented data, inconsistent entitlements, weak lineage and manual model-operating processes. **Architecture response:** Provide governed feature, document and analytical data to Vertex AI; implement MLOps and LLMOps controls; separate experimentation from production inference.

Dimension	Definition
Business outcome	Create an AI-ready and agent-ready foundation
Architecture implication	Provide governed feature, document and analytical data to Vertex AI; implement MLOps and LLMOps controls; separate experimentation from production inference.
Measures	Time to provision governed AI data; model deployment lead time; evaluation coverage; AI control compliance.
Executive accountability	Chief AI Officer, CDO and Risk
Decision consequence	Initiatives that do not map to this driver require an explicit exception or a separate approved business case.

The driver is treated as an investment constraint rather than a marketing statement. Architecture reviews must demonstrate which measurable outcome will improve, the baseline used, the target date, and the accountable owner. Benefits are tracked through the platform product backlog and quarterly value reviews.

2.9 BD-09 — Improve reliability and recoverability

Critical analytical and operational data services lack consistently tested recovery objectives and depend on manual restoration procedures. **Architecture response:** Classify workloads by service tier, define SLOs and error budgets, automate infrastructure reconstruction, retain replayable data and conduct scheduled recovery exercises.

Dimension	Definition
Business outcome	Improve reliability and recoverability
Architecture implication	Classify workloads by service tier, define SLOs and error budgets, automate infrastructure reconstruction, retain replayable data and conduct scheduled recovery exercises.
Measures	SLO attainment; mean time to restore; recovery-test pass rate; unplanned data-loss incidents.
Executive accountability	CIO, SRE Lead and Business Continuity
Decision consequence	Initiatives that do not map to this driver require an explicit exception or a separate approved business case.

The driver is treated as an investment constraint rather than a marketing statement. Architecture reviews must demonstrate which measurable outcome will improve, the baseline used, the target date, and the accountable owner. Benefits are tracked through the platform product backlog and quarterly value reviews.

2.10 BD-10 — Create transparent and sustainable economics

Shared platform costs are difficult to attribute, and teams optimize locally without visibility into enterprise unit economics. **Architecture response:** Implement labels, billing exports, reservations, showback, product-level unit metrics and a FinOps operating cadence.

Dimension	Definition
Business outcome	Create transparent and sustainable economics
Architecture implication	Implement labels, billing exports, reservations, showback, product-level unit metrics and a FinOps operating cadence.
Measures	Cost per terabyte processed; cost per product; budget variance; realized optimization savings.
Executive accountability	CFO, FinOps Lead and Platform Product Owner
Decision consequence	Initiatives that do not map to this driver require an explicit exception or a separate approved business case.

The driver is treated as an investment constraint rather than a marketing statement. Architecture reviews must demonstrate which measurable outcome will improve, the baseline used, the target date, and the accountable owner. Benefits are tracked through the platform product backlog and quarterly value reviews.

2.11 BD-11 — Increase engineering productivity

Teams repeatedly solve identity, connectivity, deployment, monitoring and pipeline patterns instead of focusing on differentiated business logic. **Architecture response:** Provide golden Terraform modules, reusable pipeline components, automated tests, paved-road CI/CD and self-service provisioning.

Dimension	Definition
Business outcome	Increase engineering productivity
Architecture implication	Provide golden Terraform modules, reusable pipeline components, automated tests, paved-road CI/CD and self-service provisioning.
Measures	Lead time for change; deployment frequency; template adoption; change-failure rate.
Executive accountability	Engineering Leadership and Platform Product Owner

Dimension	Definition
Decision consequence	Initiatives that do not map to this driver require an explicit exception or a separate approved business case.

The driver is treated as an investment constraint rather than a marketing statement. Architecture reviews must demonstrate which measurable outcome will improve, the baseline used, the target date, and the accountable owner. Benefits are tracked through the platform product backlog and quarterly value reviews.

2.12 BD-12 — Reduce architectural complexity and vendor sprawl

Acquisitions and local technology choices have produced overlapping data, integration, catalog, BI and ML platforms. **Architecture response:** Converge on a small number of strategic services, enforce exception governance and retire redundant platforms through funded migration roadmaps.

Dimension	Definition
Business outcome	Reduce architectural complexity and vendor sprawl
Architecture implication	Converge on a small number of strategic services, enforce exception governance and retire redundant platforms through funded migration roadmaps.
Measures	Strategic-service adoption; exception count and age; retired products; support-cost reduction.
Executive accountability	Enterprise Architecture Board and CIO
Decision consequence	Initiatives that do not map to this driver require an explicit exception or a separate approved business case.

The driver is treated as an investment constraint rather than a marketing statement. Architecture reviews must demonstrate which measurable outcome will improve, the baseline used, the target date, and the accountable owner. Benefits are tracked through the platform product backlog and quarterly value reviews.

2.13 Driver Prioritization and Traceability

Driver	Strategic priority	Primary capability impact	First evidence expected
BD-01	Critical	Ingestion, semantic layer, product SLOs	Pilot-domain freshness and KPI lead-time baseline
BD-02	Critical	Platform engineering, self-service, IaC	Golden path adopted by first three domains
BD-03	High	Warehouse modernization, migration factory	Validated workload waves and retirement dates
BD-04	High	Streaming, CDC, event governance	Two production streams with replay tests
BD-05	Critical	Security, privacy, audit and retention	Control mapping and restricted-data perimeter
BD-06	High	Region strategy and sovereignty	Approved residency matrix
BD-07	Critical	Data mesh and governance	Certified product catalog
BD-08	High	Vertex AI, RAG, MLOps and LLMOps	Governed AI pilot
BD-09	Critical	SRE and disaster recovery	Tiered SLO/RTO/RPO catalog
BD-10	High	FinOps and allocation	Domain showback dashboard
BD-11	High	DevSecOps and templates	Provisioning lead-time reduction
BD-12	Medium/High	Architecture governance and rationalization	Approved strategic-service catalog

3. Business and Functional Requirements

Requirements are written as verifiable enterprise obligations. Each requirement maps to an architecture response and evidence that can be reviewed during design assurance, implementation acceptance and production-readiness reviews.

Section outcomes

Prioritized requirements; Traceable architecture responses; Acceptance evidence; Named accountability.

3.1 Data Acquisition and Integration

The requirements in this category are mandatory unless the Architecture Review Board records a time-bound exception. The acceptance evidence is designed to be objectively testable rather than based on implementation claims.

ID	Requirement	Priority	Architecture response	Acceptance evidence
BR-ING-001	Ingest structured, semi-structured and unstructured data from on-premises, SaaS, cloud and partner sources.	Must	Standard ingestion patterns using Storage Transfer Service, Datastream, Pub/Sub, Dataflow, Cloud Run and approved connectors.	A representative source for each pattern passes security, reconciliation and failure-recovery testing.
BR-ING-002	Support batch, micro-batch, near-real-time and streaming delivery according to documented business latency.	Must	Latency-tiered ingestion catalog; streaming allowed only with an approved value case and SLO.	Every production feed has an approved latency class and measured end-to-end freshness.
BR-ING-003	Provide replay, backfill and idempotent reprocessing without corrupting published	Must	Immutable raw history, retained event streams, deterministic keys,	Replay exercise reproduces a defined period with zero unexplained variance.

ID	Requirement	Priority	Architecture response	Acceptance evidence
	products.		checkpoints and versioned transformations.	
BR-ING-004	Validate files, messages and records before promotion to trusted layers.	Must	Schema validation, malware/checksum controls, quarantine, dead-letter topics and quality gates.	Invalid test records are isolated with actionable failure metadata.
BR-ING-005	Reconcile source and target using counts, control totals, checksums or business balances.	Must	Automated reconciliation framework and exception workflow.	Daily reconciliation evidence meets workload-specific tolerance.
BR-ING-006	Onboard a standard source through configuration and reusable templates.	Should	Metadata-driven source registration and golden pipeline templates.	A trained domain team onboards a standard source within ten working days.

3.2 Data Management and Data Products

The requirements in this category are mandatory unless the Architecture Review Board records a time-bound exception. The acceptance evidence is designed to be objectively testable rather than based on implementation claims.

ID	Requirement	Priority	Architecture response	Acceptance evidence
BR-DAT-001	Retain an immutable source-aligned copy where legal and technical constraints allow.	Must	Cloud Storage raw zone with retention, versioning and controlled mutation permissions.	Immutability controls and restore tests are evidenced.
BR-DAT-002	Provide standardized, curated, semantic, sandbox and archive layers with distinct controls.	Must	Layered BigQuery, BigLake and Cloud Storage architecture with entry/exit criteria.	Every dataset is assigned a layer, owner, classification and lifecycle policy.
BR-DAT-003	Publish reusable data products with owners, contracts, quality objectives, SLOs and support information.	Must	Data-product manifest, Knowledge Catalog metadata and certification workflow.	First production domain publishes at least three certified products.
BR-DAT-004	Support open formats for workloads requiring cross-engine access or long-term portability.	Should	Parquet or Iceberg on Cloud Storage/BigLake where justified; native BigQuery tables for SQL-serving workloads.	Architecture decision documents format choice and performance/security validation.
BR-DAT-005	Support schema evolution without silently breaking consumers.	Must	Compatibility rules, contract versions, deprecation windows and consumer-impact analysis.	Breaking schema change is blocked unless a new version and migration plan exist.
BR-DAT-006	Implement retention, archival, legal hold and privacy deletion policies.	Must	Policy metadata, automated lifecycle actions and evidence-producing deletion workflows.	Policy tests and sampled deletion/hold evidence pass compliance review.

3.3 Governance, Metadata and Trust

The requirements in this category are mandatory unless the Architecture Review Board records a time-bound exception. The acceptance evidence is designed to be objectively testable rather than based on implementation claims.

ID	Requirement	Priority	Architecture response	Acceptance evidence
BR-GOV-001	Provide enterprise search and discovery across supported data and AI assets.	Must	Knowledge Catalog inventory, business glossary, ownership and searchable metadata.	Users locate certified products using business terminology.
BR-GOV-002	Capture technical and business lineage for material reporting and AI use cases.	Must	Automated lineage where supported plus registered custom lineage for external transformations.	Critical reports trace to authoritative sources and transformation code.
BR-GOV-003	Classify sensitive data and bind controls to classification.	Must	Sensitive Data Protection, policy tags, metadata aspects and access policies.	Restricted test data receives the expected tag and access behavior.
BR-GOV-004	Measure data quality across completeness, validity, consistency, uniqueness, timeliness and accuracy.	Must	Reusable quality rules, scorecards, certification gates and steward workflow.	Critical products meet approved thresholds for three consecutive cycles.
BR-GOV-005	Provide governed internal and external data sharing.	Must	BigQuery sharing, authorized views/datasets and API products with subscriber controls.	Consumer access is revocable, auditable and limited to contracted interfaces.
BR-GOV-006	Maintain a federated governance operating model.	Must	Central policies and standards with domain execution, RACI and automated conformance reporting.	Governance council and domain stewards approve operating procedures.

3.4 Analytics and Decision Support

The requirements in this category are mandatory unless the Architecture Review Board records a time-bound exception. The acceptance evidence is designed to be objectively testable rather than based on implementation claims.

ID	Requirement	Priority	Architecture response	Acceptance evidence
BR-ANA-001	Provide governed enterprise metrics and semantic definitions.	Must	Looker semantic models, certified BigQuery views and KPI ownership.	Executive KPIs have one approved definition and named owner.
BR-ANA-002	Enable self-service SQL analysis without unrestricted access to raw sensitive data.	Must	Curated datasets, IAM groups, policy tags, masking and sandbox guardrails.	Persona tests confirm only approved fields and rows are visible.
BR-ANA-003	Support dashboards, ad hoc exploration, spreadsheets and embedded analytics.	Should	Looker, Looker Studio, Connected Sheets and protected APIs.	Representative personas complete approved journeys within performance targets.
BR-ANA-004	Separate high-priority BI workloads from engineering and exploratory workloads.	Must	Reservations, assignments, quotas and workload-specific projects.	Concurrency test demonstrates priority workload protection.
BR-ANA-005	Provide usage, adoption and cost telemetry for analytical assets.	Should	Audit logs, INFORMATION_SCHEMA, Looker usage and product dashboards.	Monthly product review shows consumers, queries, freshness and cost.

3.5 AI and Data Science

The requirements in this category are mandatory unless the Architecture Review Board records a time-bound exception. The acceptance evidence is designed to be objectively testable rather than based on implementation claims.

ID	Requirement	Priority	Architecture response	Acceptance evidence
BR-AI-001	Provide approved, governed datasets for model training, inference and RAG.	Must	Vertex AI access to certified products, restricted service identities and lineage.	AI workload cannot bypass approved product interfaces.
BR-AI-002	Support managed experimentation, pipelines, model registry and deployment.	Must	Vertex AI Workbench, Pipelines, Model Registry and controlled endpoints.	A model progresses through development, approval and production with evidence.
BR-AI-003	Support enterprise document ingestion, embeddings, vector retrieval and grounded generation.	Should	Document AI where appropriate, embedding pipelines, Vector Search and RAG services.	Pilot provides traceable citations and passes retrieval/evaluation thresholds.
BR-AI-004	Apply responsible-AI, safety, privacy and human-oversight controls.	Must	Use-case risk tiering, evaluations, model cards, content controls and approval gates.	High-risk use case is blocked without required approvals and evidence.
BR-AI-005	Measure model, retrieval, prompt, latency, safety and token-cost performance.	Must	MLOps/LLMOps telemetry, evaluation datasets and FinOps dashboards.	Production AI service meets approved SLO and evaluation baseline.

3.6 Security, Privacy and Compliance

The requirements in this category are mandatory unless the Architecture Review Board records a time-bound exception. The acceptance evidence is designed to be objectively testable rather than based on implementation claims.

ID	Requirement	Priority	Architecture response	Acceptance evidence
BR-SEC-001	Use enterprise identities and short-lived workload credentials.	Must	Cloud Identity federation, groups, Workload Identity Federation and service-account impersonation.	No production pipeline depends on long-lived user-managed keys.
BR-SEC-002	Restrict sensitive data access by least privilege and business purpose.	Must	Project/dataset IAM, row policies, policy tags, masking and access workflows.	Quarterly access review and negative persona tests pass.
BR-SEC-003	Protect restricted data from unintended exfiltration.	Must	VPC Service Controls, Private Service Connect, controlled egress and audit monitoring.	Perimeter penetration scenarios are blocked or explicitly approved.
BR-SEC-004	Encrypt data in transit and at rest, with CMEK for selected classifications.	Must	TLS, Google default encryption and classification-based Cloud KMS policy.	Key-use, rotation and service-agent permissions pass review.
BR-SEC-005	Retain immutable audit evidence and integrate material events with the enterprise SIEM.	Must	Organization-level log sinks, protected log projects and security-event routing.	Audit-event sampling proves completeness and retention.
BR-SEC-006	Map platform controls to applicable regulatory frameworks.	Must	Reusable control catalog and workload-specific compliance overlay.	Control owners produce evidence for sampled NIST, HIPAA, GDPR, PCI and SOC 2 controls.

3.7 Platform Operations and Economics

The requirements in this category are mandatory unless the Architecture Review Board records a time-bound exception. The acceptance evidence is designed to be objectively testable rather than based on implementation claims.

ID	Requirement	Priority	Architecture response	Acceptance evidence
BR-OPS-001	Provision projects and platform capabilities through approved Infrastructure as Code.	Must	Terraform project factory and golden modules with policy checks.	No production platform resource is accepted without source-controlled definition or approved exception.
BR-OPS-002	Provide CI/CD with tests, approvals, immutable artifacts and rollback.	Must	Cloud Build and/or GitHub Actions, Artifact Registry and environment promotion.	A release demonstrates automated tests and successful rollback.
BR-OPS-003	Provide SLO-based monitoring, alerting, runbooks and ownership.	Must	Cloud Monitoring, Cloud Logging, SLO catalog and paging integration.	Each Tier 0/1 service has tested alerts and an on-call runbook.
BR-OPS-004	Allocate and optimize cost by domain, product, environment and owner.	Must	Mandatory labels/tags, billing export, reservations and showback.	At least 95% of platform spend is allocable.
BR-OPS-005	Detect configuration drift and policy non-conformance.	Must	Terraform plan, policy-as-code, asset inventory and scheduled conformance scans.	Critical drift creates a tracked incident within the defined detection SLO.
BR-OPS-006	Operate a lifecycle for service exceptions and technical debt.	Should	Architecture exception register with owner, expiry and remediation plan.	No active exception lacks an owner and review date.

3.8 Requirement Governance

- Every requirement is assigned to a business owner, architecture owner and implementation workstream before detailed design begins.
- Requirements are baselined at the end of discovery and traced to epics, controls, tests, operational procedures and acceptance evidence.
- A change affecting residency, confidentiality, recovery tier, consumer contract or material cost returns to architecture review.
- Requirements that cannot be tested are rewritten before implementation approval.
- Where regulatory obligations conflict, the stricter control applies until Legal and Privacy approve an alternative.

Acceptance rule

A capability is not considered delivered when a service is enabled. It is delivered only when the approved user journey, security behavior, failure mode, telemetry and operating ownership have been demonstrated.

4. Non-functional Requirements

Non-functional requirements define the production qualities of the platform. Targets are tiered by workload criticality and must be validated through tests, telemetry or operational exercises before go-live.

Section outcomes

Measurable service qualities; Tiered resilience targets; Security and compliance obligations; Production acceptance criteria.

4.1 Availability and Continuity

ID	Quality attribute	Target	Architecture mechanism	Evidence
NFR-AVL-001	Tier 0 data services	99.99% monthly service objective where supported by component architecture.	Multi-zone design; managed-service SLAs; dependent-service analysis.	Synthetic availability, SLO report and dependency test.
NFR-AVL-002	Tier 1 data services	99.9% monthly service objective.	Resilient regional design and tested restart procedures.	Monthly SLO report.
NFR-AVL-003	Planned maintenance	No uncommunicated outage for Tier 0/1 products.	Maintenance windows, consumer notices and backward-compatible releases.	Change records and consumer notification evidence.
NFR-AVL-004	Dependency degradation	Critical products must fail safely and expose degraded-state telemetry.	Timeout, retry, circuit-breaker and fallback design where applicable.	Failure-injection test.
NFR-AVL-005	Control-plane outage	Existing data serving should continue where service characteristics permit.	Decouple runtime from deployment tooling; cache approved configuration.	Control-plane failure scenario review.

Targets are baseline enterprise objectives. A product may adopt stronger targets when business impact warrants the additional cost, or weaker targets only when its service tier, consumer contract and risk acceptance explicitly permit them.

4.2 Recovery and Data Protection

ID	Quality attribute	Target	Architecture mechanism	Evidence
NFR-DR-001	Tier 0 recovery	RTO <1 hour; RPO <5 minutes.	Replayable events, multi-region runbook, IaC reconstruction and data recovery pattern.	Annual full failover and quarterly component tests.
NFR-DR-002	Tier 1 recovery	RTO <4 hours; RPO <15 minutes.	Regional recovery environment and automated configuration restore.	Semiannual exercise.
NFR-DR-003	Tier 2 recovery	RTO <24 hours; RPO <24 hours.	Backup/restore and prioritized reconstruction.	Annual exercise.
NFR-DR-004	Data corruption	Point-in-time recovery or source replay for governed critical datasets.	BigQuery time travel/snapshots, versioned objects, immutable raw and reconciliation.	Corruption simulation.
NFR-DR-005	Cyber recovery	Protected recovery copies and clean-room procedures for designated workloads.	Separated identities, immutable evidence and staged restoration.	Tabletop and technical recovery test.

Targets are baseline enterprise objectives. A product may adopt stronger targets when business impact warrants the additional cost, or weaker targets only when its service tier, consumer contract and risk acceptance explicitly permit them.

4.3 Performance and Latency

ID	Quality attribute	Target	Architecture mechanism	Evidence
NFR-PERF-001	Streaming freshness	95th percentile source-to-curated latency ≤60 seconds for approved real-time products.	Pub/Sub/Dataflow capacity model and end-to-end timestamps.	Peak-load test.
NFR-PERF-002	Near-real-time freshness	95th percentile latency ≤5 minutes.	CDC or micro-batch pattern.	Sustained-volume test.
NFR-PERF-003	Batch completion	Critical daily products complete before 06:00 local business time.	Dependency-aware orchestration and capacity isolation.	Three-cycle operational proof.
NFR-PERF-004	Interactive BI	95th percentile certified dashboard response ≤5 seconds for agreed concurrency.	Model optimization, aggregate strategy, BI Engine where justified.	Concurrency test.
NFR-PERF-005	Ad hoc analytics	95th percentile standard governed query ≤30 seconds for documented data volume.	Partitioning, clustering, reservation and query standards.	Representative benchmark suite.

Targets are baseline enterprise objectives. A product may adopt stronger targets when business impact warrants the additional cost, or weaker targets only when its service tier, consumer contract and risk acceptance explicitly permit them.

4.4 Scalability and Capacity

ID	Quality attribute	Target	Architecture mechanism	Evidence
NFR-SCL-001	Ingestion growth	Scale from 50 TB/day baseline to 250 TB/day without architectural redesign.	Autoscaling services, quota planning and partitioned pipelines.	5x volume test or validated capacity model.
NFR-SCL-002	Streaming bursts	Absorb 3x average event rate for 30 minutes without data loss.	Pub/Sub retention, Dataflow autoscaling and backlog monitoring.	Burst test.
NFR-SCL-003	User growth	Support 10,000 analytical users and 1,000 concurrent users through workload isolation.	Reservations, semantic caching and concurrency design.	Load model and phased tests.
NFR-SCL-004	Domain growth	Onboard at least 50 domains using repeatable project and product patterns.	Project factory and standardized modules.	Provisioning demonstration.
NFR-SCL-005	Metadata scale	Catalog millions of assets without manual registration processes.	Automated harvesting and metadata APIs.	Catalog-load test and search sampling.

Targets are baseline enterprise objectives. A product may adopt stronger targets when business impact warrants the additional cost, or weaker targets only when its service tier, consumer contract and risk acceptance explicitly permit them.

4.5 Security and Privacy

ID	Quality attribute	Target	Architecture mechanism	Evidence
NFR-SEC-001	Authentication	All human access uses federated identity and phishing-resistant MFA where required.	Identity federation and conditional access.	Access-control test.
NFR-SEC-002	Privilege	No standing owner/editor access in production; privileged roles are time-bound.	PAM/JIT, group controls and denial policies.	Quarterly entitlement review.
NFR-SEC-003	Credential hygiene	No user-managed service-account keys for production workloads.	Workload Identity Federation and impersonation.	Continuous key scan.
NFR-SEC-004	Sensitive data	Restricted data is classified, protected, logged and excluded from unauthorized lower environments.	Sensitive Data Protection, policy tags, masking and perimeter controls.	Positive and negative persona tests.
NFR-SEC-005	Detection	Critical security events are delivered to SOC tooling within five minutes.	Central sinks and event routing.	Detection pipeline test.

Targets are baseline enterprise objectives. A product may adopt stronger targets when business impact warrants the additional cost, or weaker targets only when its service tier, consumer contract and risk acceptance explicitly permit them.

4.6 Data Quality and Trust

ID	Quality attribute	Target	Architecture mechanism	Evidence
NFR-DQ-001	Completeness	Critical key fields meet ≥99.5% completeness unless product contract states otherwise.	Automated rules and publishing gate.	Three-cycle scorecard.
NFR-DQ-002	Validity	Critical conforming fields meet ≥99.5% rule validity.	Reference and pattern validation.	Quality report.
NFR-DQ-003	Freshness	≥99% of Tier 0/1 products meet their freshness SLO each month.	End-to-end timestamps and alerting.	Monthly SLO report.
NFR-DQ-004	Reconciliation	Financial/regulatory products reconcile to source control totals at 100% or approved tolerance.	Control-total framework and exception workflow.	Signed reconciliation evidence.
NFR-DQ-005	Lineage	100% of critical products expose source-to-consumer lineage at the documented transformation boundary.	Automated and registered lineage.	Lineage audit.

Targets are baseline enterprise objectives. A product may adopt stronger targets when business impact warrants the additional cost, or weaker targets only when its service tier, consumer contract and risk acceptance explicitly permit them.

4.7 Operability and Observability

ID	Quality attribute	Target	Architecture mechanism	Evidence
NFR-OPS-001	Telemetry	Every production service emits logs, metrics and ownership metadata.	Standard modules and observability library.	Production-readiness review.
NFR-OPS-002	Alert quality	Alerts are actionable, severity-mapped and linked to runbooks.	Alert standards and SLO-based paging.	Alert review and drill.
NFR-OPS-003	Incident response	Tier 0 incidents acknowledged within five minutes and assigned an incident commander.	On-call model and incident tooling integration.	Incident exercise.
NFR-OPS-004	Change traceability	Every production change maps to source commit, build, approver and deployment record.	CI/CD provenance and audit logs.	Release sample.
NFR-OPS-005	Operational readiness	No Tier 0/1 product goes live without ownership, SLO, dashboard, alerts, runbook and recovery test.	Mandatory readiness gate.	Signed checklist.

Targets are baseline enterprise objectives. A product may adopt stronger targets when business impact warrants the additional cost, or weaker targets only when its service tier, consumer contract and risk acceptance explicitly permit them.

4.8 Maintainability and Engineering Quality

ID	Quality attribute	Target	Architecture mechanism	Evidence
NFR-MNT-001	Infrastructure automation	≥95% of managed platform resources are defined through approved IaC.	Terraform modules and exceptions register.	Asset-to-code reconciliation.
NFR-MNT-002	Test automation	Critical pipelines include unit, contract, quality, integration and deployment tests.	Reusable test harness.	Pipeline evidence.
NFR-MNT-003	Versioning	Schemas, transformations, infrastructure and data-product contracts are version controlled.	Git-based workflow and release tags.	Repository audit.
NFR-MNT-004	Recovery from change	Rollback or forward-fix procedure exists for every deployment type.	Immutable artifacts and database/schema migration strategy.	Rollback drill.
NFR-MNT-005	Documentation	Architecture, runbooks and product metadata are updated in the same change lifecycle.	Docs-as-code and metadata checks.	Release audit.

Targets are baseline enterprise objectives. A product may adopt stronger targets when business impact warrants the additional cost, or weaker targets only when its service tier, consumer contract and risk acceptance explicitly permit them.

4.9 Cost and Sustainability

ID	Quality attribute	Target	Architecture mechanism	Evidence
NFR-CST-001	Allocation	≥95% of monthly platform spend is attributable to domain, product, environment and owner.	Labels/tags, reservations and billing export.	FinOps report.
NFR-CST-002	Budget control	Forecast variance remains within ±10% at portfolio level after stabilization.	Forecasting and anomaly alerts.	Monthly review.
NFR-CST-003	Query efficiency	Curated workloads demonstrate partition pruning and avoid unrestricted scans.	SQL linting, maximum bytes and workload tests.	Query-plan review.
NFR-CST-004	Idle resources	Ephemeral compute terminates automatically and idle non-production capacity is minimized.	Serverless services, schedules and lifecycle automation.	Idle-resource report.
NFR-CST-005	Sustainability	Region, retention and processing choices consider carbon and resource efficiency without violating residency or resilience.	Architecture trade-off record.	Quarterly architecture review.

Targets are baseline enterprise objectives. A product may adopt stronger targets when business impact warrants the additional cost, or weaker targets only when its service tier, consumer contract and risk acceptance explicitly permit them.

4.10 Compliance and Auditability

ID	Quality attribute	Target	Architecture mechanism	Evidence
NFR-CMP-001	Evidence	Control evidence is reproducible from configuration, logs and workflow records.	Automated evidence collection.	Audit sample.
NFR-CMP-002	Retention	Data and logs follow approved retention schedules with legal-hold overrides.	Lifecycle policy and hold workflow.	Policy test.
NFR-CMP-003	Privacy deletion	Approved deletion requests propagate to governed copies within the policy timeframe.	Identity mapping, delete workflow and evidence.	End-to-end deletion test.
NFR-CMP-004	Residency	Restricted data remains within approved locations unless a transfer is authorized.	Resource-location policies and transfer controls.	Asset scan.
NFR-CMP-005	Control exceptions	Every exception has owner, rationale, compensating control, expiry and review date.	Exception register.	Quarterly review.

Targets are baseline enterprise objectives. A product may adopt stronger targets when business impact warrants the additional cost, or weaker targets only when its service tier, consumer contract and risk acceptance explicitly permit them.

4.11 Service-tier Model

Tier	Business impact	Availability objective	RTO	RPO	Required assurance
Tier 0	Immediate material financial, safety, regulatory or customer impact	99.99% target	<1 hour	<5 minutes	24×7 on-call; quarterly recovery component test; annual regional exercise

Tier	Business impact	Availability objective	RTO	RPO	Required assurance
Tier 1	Major business-process disruption or missed regulatory/operational deadline	99.9% target	<4 hours	<15 minutes	24x7 or business-critical on-call; semiannual recovery exercise
Tier 2	Important analytics with tolerable delayed recovery	99.5% target	<24 hours	<24 hours	Business-hours support; annual restore test
Tier 3	Experimentation, development or disposable workloads	Best effort	Rebuild	No guarantee	Quota, expiry and data-class restrictions

Verification policy

Availability, latency, quality and recovery objectives must be measured at the data-product or user-journey boundary. Provider service availability alone is not sufficient evidence.

5. Architecture Principles

The principles are binding design constraints for the enterprise platform. They are intended to reduce one-off decisions, improve interoperability and make security, reliability and economics repeatable across domains.

Section outcomes

Twenty enforceable principles; Required behaviors; Exception criteria; Conformance evidence.

5.1 AP-01 — Business value before technology

Every platform capability and real-time requirement must map to a measurable business, regulatory or operational outcome.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Architecture proposals state the outcome, baseline, target and owner; unnecessary low-latency or duplicated capabilities are rejected.
Exception rule	A separate approved risk or regulatory obligation justifies the capability.
Conformance evidence	Business case, outcome metric and portfolio traceability.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.2 AP-02 — Data is an enterprise asset and a domain-owned product

Domains own meaning, quality and support; the platform provides shared controls and delivery mechanisms.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Each published product has an owner, contract, SLO, classification, consumers, lifecycle state and support model.
Exception rule	Temporary migration datasets may have transitional ownership with an expiry date.
Conformance evidence	Catalog metadata, product manifest and RACI.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.3 AP-03 — Security and privacy by design

Security controls are designed with the service, not added after implementation.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Threat modeling, least privilege, data classification, private access, logging and privacy controls are part of the initial design.
Exception rule	No exception for legal or regulated controls; technical exceptions need compensating controls and expiry.
Conformance evidence	Threat model, access tests, policy scans and security review.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.4 AP-04 — Zero Trust access

Trust is based on verified identity, context, device/workload posture and explicit authorization rather than network location.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Federated identities, short-lived credentials, service-account impersonation and context-aware access are used.
Exception rule	Legacy protocols may use isolated gateways during migration.
Conformance evidence	Identity-flow diagram, entitlement review and credential scan.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.5 AP-05 — Regional sovereignty with global governance

Data is processed in approved locations while policies, metadata and product standards remain globally consistent.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Residency classes determine project, region, transfer and replication behavior.
Exception rule	Cross-region transfer requires legal purpose, data-owner approval and recorded controls.
Conformance evidence	Residency matrix, resource-location policy and transfer register.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.6 AP-06 — Managed and serverless first

Use managed and serverless services when they satisfy functional, security, performance and portability requirements.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	BigQuery, Dataflow, Pub/Sub, Datastream, Cloud Run and managed orchestration are preferred to self-managed clusters.
Exception rule	Self-managed technology requires a documented feature gap, support model and exit plan.
Conformance evidence	Technology-selection matrix and ADR.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.7 AP-07 — Separate storage and compute

Persistent data is not coupled to fixed processing capacity.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Use Cloud Storage, BigQuery and BigLake as durable layers; allocate independent compute for SQL, streaming, Spark, BI and AI.
Exception rule	Specialized operational stores may combine characteristics when justified.
Conformance evidence	Physical architecture and cost model.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.8 AP-08 — Open formats where they create value

Open formats are used for interoperability, archival or multi-engine access—not as a blanket rule.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Parquet or Iceberg is considered for lakehouse workloads; native BigQuery tables are preferred for managed analytical serving.
Exception rule	A closed format may be used where it materially improves reliability, security or service capability.
Conformance evidence	Format decision, interoperability test and lifecycle plan.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.9 AP-09 — Immutable source history and reproducible processing

The enterprise must be able to explain and reproduce published data.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Retain replayable raw records or equivalent source evidence; transformations are versioned and deterministic where practical.
Exception rule	Legal deletion and source restrictions may require controlled mutation.
Conformance evidence	Replay test, code version and reconciliation evidence.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.10 AP-10 — Metadata and contracts are first-class

Schemas, ownership, quality, SLOs, classifications and interfaces are managed as deployable metadata.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Pipelines cannot publish certified products without contract metadata and compatibility checks.
Exception rule	Exploratory assets may use lightweight metadata but must be clearly non-certified.
Conformance evidence	Catalog checks and CI validation.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.11 AP-11 — Automate policy and delivery

Manual console configuration is not the production operating model.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Use Terraform, policy-as-code, CI/CD, automated quality gates and evidence collection.
Exception rule	Emergency changes are time-bound, logged and reconciled to code.
Conformance evidence	Asset-to-code reconciliation and deployment provenance.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.12 AP-12 — Design for failure and recovery

Every critical data flow has documented failure modes, recovery procedures and tested objectives.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Include retries, dead letters, checkpoints, replay, backup/restore, dependency failure and regional recovery.
Exception rule	Tier 3 experiments may be rebuild-only.
Conformance evidence	Failure-mode analysis, runbook and test evidence.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.13 AP-13 — Observability at the product boundary

Technical component health is insufficient without freshness, quality and consumer-impact telemetry.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Measure end-to-end latency, product availability, quality, adoption, cost and dependencies.
Exception rule	None for Tier 0/1 products; lightweight telemetry permitted for sandboxes.
Conformance evidence	SLO dashboard, alert tests and owner metadata.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.14 AP-14 — Cost is an architecture metric

Teams own the economic consequences of design and consumption.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Allocate cost by product/domain, define unit economics, use budgets, reservations and lifecycle policies, and review anomalies.
Exception rule	Shared foundational spend may be centrally funded but remains transparent.
Conformance evidence	Showback report, unit metrics and optimization backlog.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.15 AP-15 — Paved roads over unrestricted choice

The platform offers approved golden paths that cover common workloads and accelerate delivery.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Teams start with reusable modules, pipeline templates, IAM patterns and operating standards.
Exception rule	Exceptions require a feature gap, risk analysis, ownership and retirement/standardization plan.
Conformance evidence	Template adoption and exception register.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.16 AP-16 — Least data movement

Move or copy data only for residency, performance, isolation, recovery or contractual reasons.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Prefer in-place governed access, authorized views and managed sharing; avoid uncontrolled extracts.
Exception rule	Approved regulatory archive or external contract may require a copy.
Conformance evidence	Data-flow inventory and transfer-cost analysis.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.17 AP-17 — Consumer contracts and backward compatibility

Published products evolve without silently breaking consumers.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Use semantic versioning, compatibility tests, deprecation windows and migration communications.
Exception rule	Emergency security fixes may require accelerated change with executive communication.
Conformance evidence	Contract test and consumer-impact record.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.18 AP-18 — Federated governance, centrally assured

Central teams define policies and guardrails; domains execute and own outcomes.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Automated conformance, common classifications and delegated stewardship are required.
Exception rule	Highly regulated domains may use additional centralized approval.
Conformance evidence	Governance RACI and conformance dashboard.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.19 AP-19 — AI uses governed data and explicit risk tiers

AI does not create an alternative uncontrolled path to enterprise information.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Training, RAG and inference use approved data products, service identities, evaluations, audit and human oversight based on risk.
Exception rule	Public low-risk experimentation may use approved non-sensitive data in sandbox controls.
Conformance evidence	AI use-case registration, evaluation report and model card.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.20 AP-20 — Prefer reversible decisions and planned exits

Architecture minimizes lock-in where it is economically sensible and documents exit strategies for material dependencies.

This principle applies to new platform capabilities, domain data products and material changes to existing services. Reviewers should assess both the design intent and the evidence produced by implementation.

Element	Requirement
Required behavior	Use interfaces, portable data formats, versioned code, export procedures and periodic decision reviews.
Exception rule	Deep managed-service adoption is accepted when benefits exceed switching risk.
Conformance evidence	ADR review trigger, recovery/export test and lifecycle plan.
Review owner	Enterprise Architecture with Security, Data Governance, SRE or FinOps participation as applicable.
Failure consequence	The design remains conditional or is recorded as a time-bound architecture exception.

5.21 Principle Conformance Model

Rating	Definition	Action
Conformant	Design and evidence satisfy the principle.	Approve within normal governance.
Conditionally conformant	Intent is met but evidence or automation is incomplete.	Approve with dated conditions and accountable owner.
Exception	Design intentionally departs from the principle.	Architecture exception with risk, compensating control and expiry.
Non-conformant	Departure is unacknowledged or creates unacceptable risk.	Do not approve.

6. Cloud Adoption Strategy

The adoption strategy treats the data platform as an enterprise product and transformation program, not as a collection of migrations. It establishes how foundations, domains, controls, skills, economics and legacy retirement progress together.

Section outcomes

Adoption waves; Workstream responsibilities; Migration decision model; Value and risk controls.

6.1 Adoption Thesis

The enterprise should establish a secure, reusable cloud foundation and prove the operating model with a bounded production domain before accelerating workload migration. The sequence avoids two common failure modes: building a technically complete platform without product adoption, and migrating workloads rapidly into inconsistent projects that later require expensive remediation.

Adoption position

Build the minimum viable foundation required for the first production product, then expand platform capabilities through repeated domain adoption. Do not wait for an exhaustive platform before delivering value, and do not permit production shortcuts that undermine identity, network, audit, residency or recovery controls.

6.2 Adoption Workstreams

ID	Workstream	Scope	Accountable lead
WS-01	Executive sponsorship and value	Outcome portfolio, funding model, business ownership, decision escalation.	Executive sponsor, CDO, CFO
WS-02	Cloud foundation	Organization, folders, projects, Shared VPC, identity, logging, KMS and policy.	Cloud Platform Lead
WS-03	Security and compliance	Threat model, service perimeter, controls, evidence, privacy and risk acceptance.	CISO and Compliance
WS-04	Data platform product	Ingestion, storage, processing, governance, sharing, templates and product backlog.	Data Platform Product Owner
WS-05	Domain adoption	Source onboarding, contracts, quality, product delivery and consumer migration.	Domain Data Product Owners

ID	Workstream	Scope	Accountable lead
WS-06	Migration factory	Assessment, code conversion, reconciliation, cutover, coexistence and decommissioning.	Migration Lead
WS-07	Analytics and AI	Semantic layer, governed BI, AI data access, MLOps and RAG patterns.	Analytics/AI Leads
WS-08	SRE and operations	SLOs, observability, incident, capacity, recovery, support and readiness.	SRE Lead
WS-09	DevSecOps and IaC	Repositories, golden modules, build, testing, provenance and promotion.	DevSecOps Lead
WS-10	FinOps	Allocation, forecasts, reservations, cost controls and unit economics.	FinOps Lead
WS-11	Change and skills	Role design, training, communities of practice and adoption communications.	Transformation Lead
WS-12	Architecture governance	ADRs, standards, exceptions, assurance and technology lifecycle.	Chief Architect

6.3 Migration Disposition Model

Each workload is assessed using business criticality, strategic differentiation, technical debt, data sensitivity, latency, integration complexity, regulatory constraints, performance profile and retirement dependency. The selected disposition must include a target date, funding owner and measurable exit criterion.

Disposition	When to use	Typical implementation	Exit criterion
Retire	Unused, duplicated or legally obsolete capability.	Archive required evidence and remove integrations.	No active consumers; retention obligations satisfied.
Retain	Migration risk or dependency exceeds current value.	Keep in place with interface and control improvements.	Scheduled review date and funded future decision.
Replace	Commodity capability is better served by strategic SaaS or managed product.	Migrate consumers and history to approved replacement.	Old product disabled and support contract ended.
Rehost	Short-term infrastructure exit with minimal application change.	Move compute/storage while preserving interfaces.	Data center dependency removed; modernization debt recorded.
Replatform	Managed cloud service satisfies requirements with moderate change.	Move warehouse/pipeline to BigQuery, Dataflow or managed equivalent.	Performance, reconciliation, security and operations accepted.
Refactor	Differentiated capability requires cloud-native redesign.	Rebuild around data products, events, APIs or AI.	Business outcomes and SLOs met; legacy service retired.
Consolidate	Multiple regional or departmental implementations perform the same function.	Adopt strategic platform service and migrate variants.	Exceptions retired and shared capability meets demand.

6.4 Workload Prioritization

Factor	Weight	Scoring guidance
Business value	25%	Revenue, risk reduction, customer impact, regulatory deadline or operating efficiency.
Platform learning	15%	Exercises reusable patterns without creating unbounded complexity.

Factor	Weight	Scoring guidance
Data readiness	15%	Known ownership, source access, quality baseline and consumer commitment.
Technical feasibility	15%	Supported interfaces, manageable transformation and achievable non-functional targets.
Risk and compliance	15%	Removes control gaps or meets an external commitment.
Legacy cost avoidance	10%	Enables license, infrastructure or support retirement.
Organizational readiness	5%	Named product owner, engineering capacity and change plan.

The first domain should score highly on business value and learning while remaining bounded enough to complete. A Customer 360, regulatory reporting or supply-chain visibility product is typically preferable to an enterprise-wide master-data replacement as the initial production scope.

6.5 Adoption Waves

Wave	Indicative timing	Scope	Exit evidence
Wave 0 — Mobilize	0–8 weeks	Confirm sponsorship, assumptions, control scope, platform product charter, architecture baseline and prioritized pilot.	Approved charter; decision rights; baseline requirements; funded team.
Wave 1 — Establish foundation	6–16 weeks	Deploy organization/folder/project patterns, Shared VPC, identity, KMS, central logs, CI/CD and policy guardrails.	Foundation acceptance and security assurance.
Wave 2 — Minimum viable platform	10–24 weeks	Implement landing/raw/curated patterns, batch/CDC/streaming templates, governance, quality, monitoring and FinOps.	Reference pipelines and first certified product.
Wave 3 — Prove operating model	16–32 weeks	Run the first domain in production, prove product ownership, SLOs, support, recovery and cost review.	Three months of stable product operation.
Wave 4 — Scale domains	6–24 months	Onboard priority domains through repeatable waves; add capabilities driven by validated demand.	Domain adoption velocity and reuse targets.
Wave 5 — Migrate and retire	12–36 months	Move legacy warehouses, Hadoop, ETL and BI; reconcile and decommission.	Contract/license exits and infrastructure shutdown.
Wave 6 — Optimize and innovate	Continuous	Tune cost/performance, automate governance, expand AI/agents and improve product experience.	Quarterly value, risk and maturity improvement.

6.6 Coexistence and Transition Architecture

- Maintain an authoritative-source register during coexistence; avoid two systems both being treated as the system of record.
- Use controlled replication and reconciliation rather than unmanaged extracts.
- Freeze nonessential legacy enhancements after a migration wave is committed.
- Publish consumer migration dates and provide compatibility views only for a defined deprecation period.
- Measure duplicate run cost and operational risk as explicit transformation debt.
- Decommission only after retention, audit, legal hold, recovery and consumer obligations are confirmed.

6.7 Skills and Operating-model Transition

Audience	Required capability	Enablement mechanism	Proficiency evidence
Platform engineers	Terraform, networking, IAM, KMS, service perimeters and platform APIs.	Hands-on labs, paired delivery and module ownership.	Can deploy and troubleshoot a governed project pattern.
Data engineers	Beam/Dataflow, BigQuery, Dataform, orchestration, quality and contracts.	Reference implementations and code reviews.	Delivers a compliant pipeline using golden templates.
Domain product teams	Product ownership, SLOs, metadata, quality and consumer management.	Product coaching and domain playbook.	Publishes and operates a certified product.
Security and risk	Cloud control model, evidence, service behavior and threat analysis.	Control workshops and assurance exercises.	Approves workload based on automated evidence.
SRE/operations	Cloud telemetry, incident, capacity, error budgets and recovery.	Game days and production-readiness reviews.	Runs incident and restore exercises.
FinOps and finance	Cloud allocation, reservations, unit economics and forecasting.	Dashboard workshops and monthly review.	Explains spend variance and action plan.
Executives	Decision rights, product funding and transformation outcomes.	Quarterly steering reviews.	Resolves cross-domain ownership and funding issues.

6.8 Funding and Product Management

Foundational capabilities should be funded as an enterprise platform product with a multi-quarter roadmap. Domain-specific products are funded by the business outcomes they support. Shared costs remain transparent; showback precedes chargeback until allocation rules are stable and avoid discouraging adoption through premature internal pricing.

Funding object	Funding approach	Decision cadence	Key metric
Enterprise foundation	Central strategic investment	Annual funding with quarterly reprioritization	Control coverage and platform availability
Platform capabilities	Product-based capacity and roadmap funding	Quarterly	Adoption, lead time, reliability and cost per onboarded product
Domain data products	Business-value funding	Quarterly product review	Outcome, SLO, quality, adoption and unit cost
Migration waves	Program funding tied to retirement benefits	Wave gate	Reconciliation, cutover and cost avoided
Innovation/AI experiments	Time-bound portfolio funding	Stage gate	Evaluation result and path to production or closure

6.9 Adoption Risks and Controls

Risk	Early indicator	Preventive control	Corrective action
Foundation over-engineering	Long design period without a production consumer.	Anchor scope to first-domain needs and non-negotiable controls.	Time-box decisions and defer optional services.
Migration without retirement	Cloud and legacy costs both rise.	Fund decommissioning and define exit criteria per wave.	Escalate unresolved consumer dependencies.
Central team bottleneck	Domain backlog grows and local workarounds appear.	Self-service modules and delegated ownership.	Increase platform capacity and simplify approval paths.
Uncontrolled decentralization	Inconsistent projects, identities and services.	Project factory, policy-as-code and conformance reporting.	Contain, remediate and time-bound exceptions.
AI experimentation bypasses governance	Sensitive data appears in unmanaged notebooks or prompts.	Approved AI sandboxes and governed data interfaces.	Suspend access, investigate and strengthen controls.
Skills lag behind delivery	High defect rate and consultant dependence.	Role-based enablement and paired engineering.	Reduce wave concurrency and invest in internal ownership.

6.10 Adoption Success Measures

Dimension	Year 1 target	Year 2 target	Measurement
Time to provision governed domain environment	≤10 working days	≤2 working days	Project-factory telemetry
Standard source onboarding	≤20 working days	≤10 working days	Pipeline delivery metrics
Golden-template adoption	≥70%	≥90%	Repository and deployment metadata
Certified data products	10–20	50+	Catalog lifecycle state
Platform SLO attainment	≥99.9% Tier 1	Sustained; Tier 0 where approved	SLO dashboard
Spend allocation	≥90%	≥95%	Billing export and metadata
Legacy workloads retired	Wave-specific	Portfolio majority	Application portfolio record
Recovery exercises passed	All Tier 0/1 pilot services	All Tier 0/1 services	DR evidence repository

7. Enterprise Reference Architecture

The reference architecture defines the reusable target state for enterprise data, analytics and AI workloads. It separates business capabilities from logical planes and physical deployment so teams can adopt consistent controls without forcing every workload into an identical implementation.

Section outcomes

Conceptual, logical and physical views; Responsibility boundaries; Primary data flows; Strategic service profiles.

7.1 Conceptual Reference Architecture

The conceptual view organizes the platform around producers, integration, durable data, processing, governance, consumption and operations. Security, reliability, metadata and FinOps are cross-cutting control planes rather than independent afterthoughts.

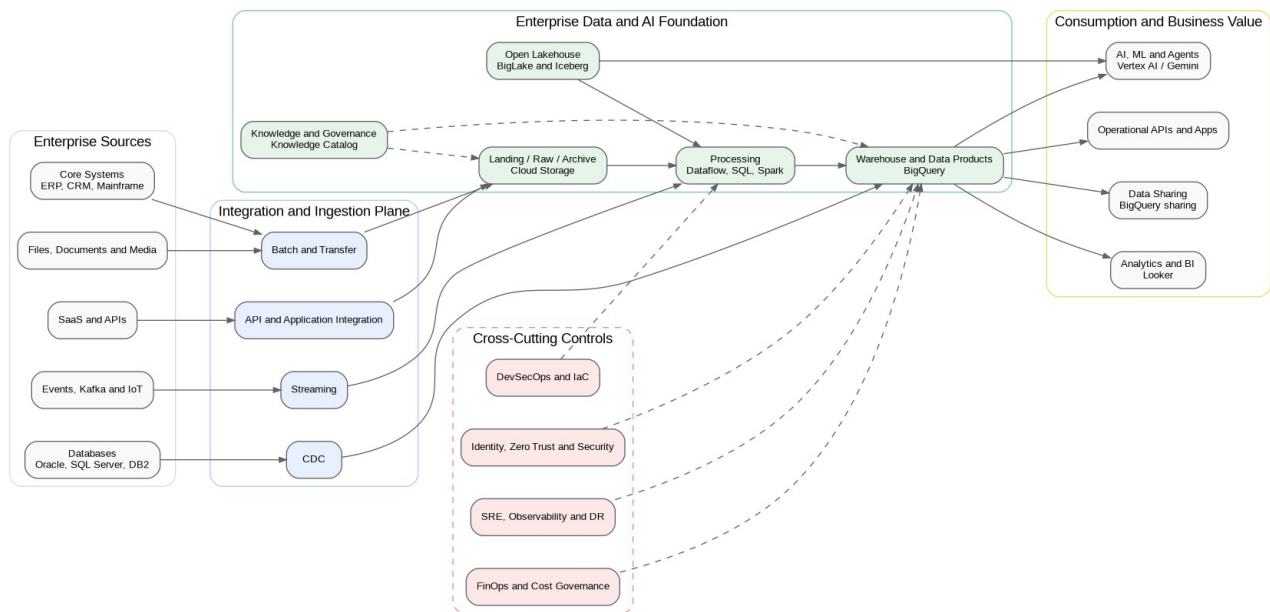


Figure 7-1. Enterprise conceptual architecture showing producers, platform planes, consumers and cross-cutting control functions.

Conceptual capability	Responsibilities	Primary outcomes
Data producers	Operational systems, SaaS, files, partners, events, documents and cross-cloud sources.	Authoritative source ownership and controlled interfaces.
Integration plane	Batch transfer, CDC, events, APIs, validation, quarantine and reconciliation.	Reliable and traceable movement into the platform.
Data foundation	Immutable raw history, standardized data, curated products, semantic assets and archive.	Durable, governed and reusable enterprise information.
Processing plane	SQL, batch, streaming, Spark, orchestration and metadata-driven execution.	Scalable transformation with repeatability and failure recovery.
Governance plane	Catalog, glossary, lineage, quality, contracts, classification and certification.	Discoverable and trusted data products.
Consumption plane	BI, SQL, APIs, sharing, operational applications, data science and AI.	Controlled access to fit-for-purpose interfaces.
Platform control plane	Identity, network, encryption, policy, CI/CD, observability, SRE and FinOps.	Consistent security, operations and economics.

7.2 Logical Reference Architecture

The logical view establishes separation between the data plane that carries business information and the control planes that configure, protect, observe and govern it. This separation supports federated domain ownership while preserving enterprise assurance.

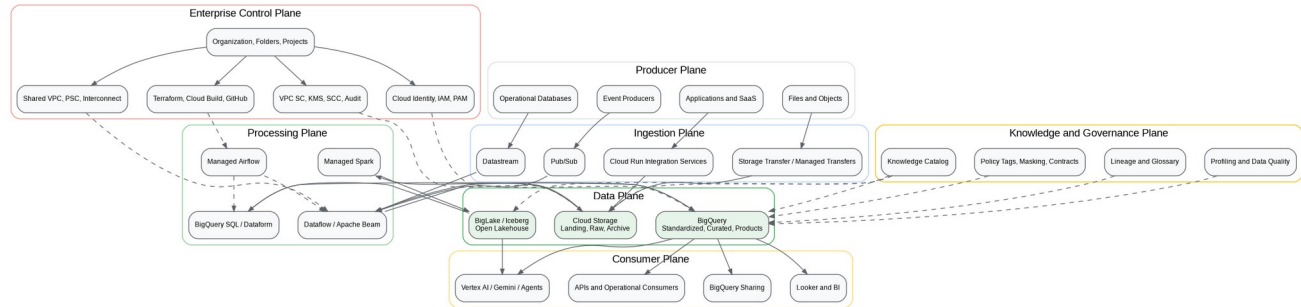


Figure 7-2. Logical reference architecture with ingestion, data, processing, governance, security, delivery and operations planes.

Plane	Responsibility	Representative services
Ingestion plane	Accepts files, database changes, messages, APIs and bulk transfers; validates envelope and schema; records source, batch and event identity.	Storage Transfer Service, Datastream, Pub/Sub, Dataflow, Cloud Run, approved connectors.
Data plane	Stores immutable source history, standardized entities, curated products, semantic models, feature/document assets and archives.	Cloud Storage, BigQuery, BigLake; specialized stores only when justified.
Processing plane	Executes deterministic transformations, stream processing, SQL models, Spark jobs, orchestration and quality rules.	Dataflow, BigQuery SQL, Dataform, Serverless for Apache Spark, Managed Service for Apache Airflow, Workflows.
Governance plane	Registers assets, ownership, contracts, classification, glossary, lineage, quality and lifecycle state.	Knowledge Catalog, Sensitive Data Protection, BigQuery governance features and product metadata.
Security plane	Controls human/workload identity, network paths, service perimeters, keys, secrets and audit.	Cloud IAM, Workforce/Workload Identity Federation, Shared VPC, PSC, VPC SC, Cloud KMS, Secret Manager, SCC.
Delivery plane	Presents semantic, SQL, shared, API and AI-ready interfaces to approved consumers.	Looker, BigQuery, BigQuery sharing, Cloud Run/API management, Vertex AI.
Operations plane	Measures technical and product health, controls delivery, allocates cost and recovers services.	Cloud Monitoring, Cloud Logging, Terraform, Cloud Build/GitHub Actions, billing export, SRE processes.

7.3 Physical Global Topology

The physical topology uses globally consistent foundations with regional data-residency zones. A region pair is selected per legal and business geography after validating service availability, inter-region behavior, recovery objectives and transfer restrictions. The diagram is a pattern, not a claim that every service automatically provides cross-region failover.

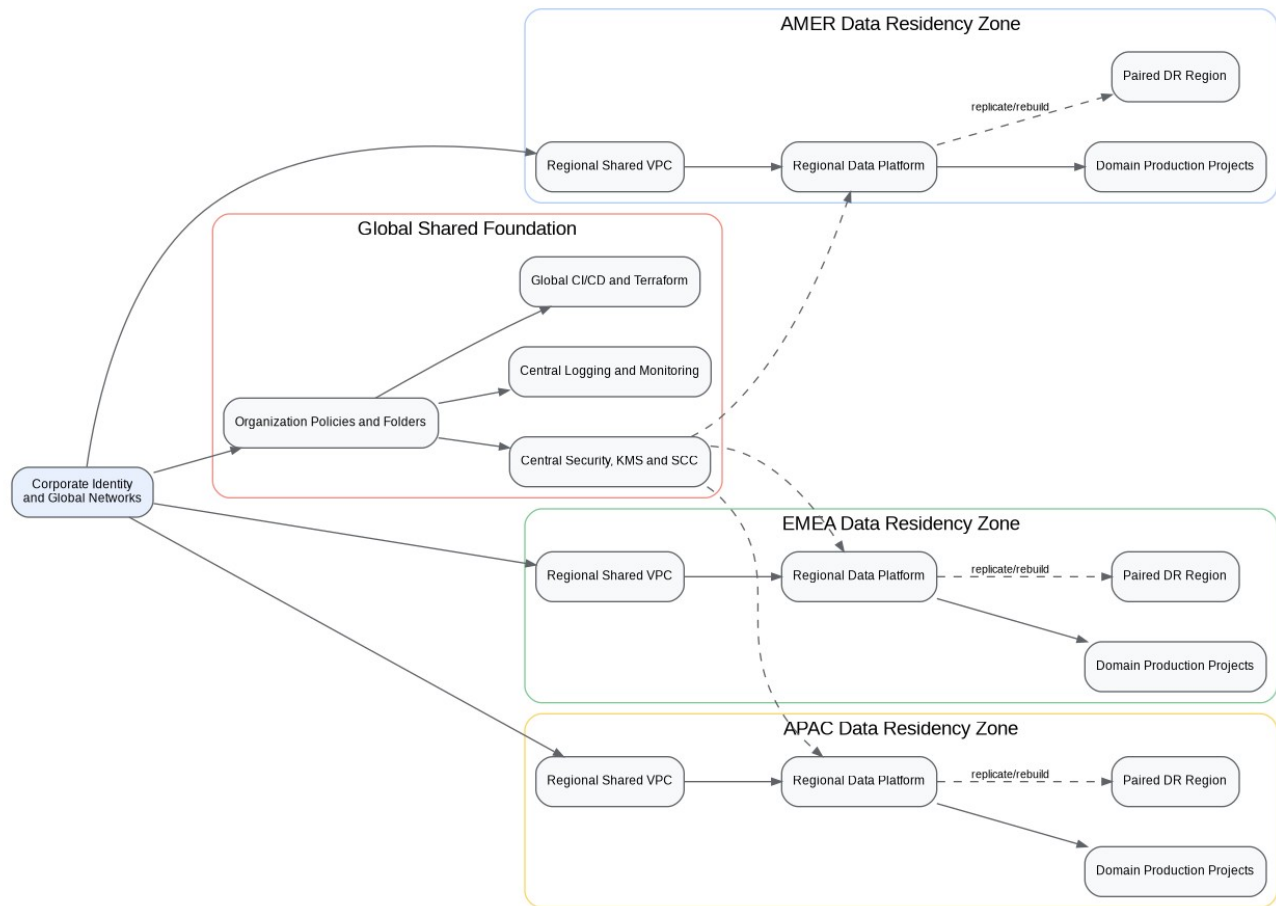


Figure 7-3. Global topology with enterprise foundation, regional data zones and controlled inter-region exchange.

Layer	Physical design	Key control
Enterprise organization	Central organization, policy hierarchy, security and shared-services folders.	Organization policy, billing, central log sinks and delegated administration.
Global shared services	Identity federation, source control, artifact repositories, policy pipelines, catalog standards and enterprise monitoring.	Global control consistency with restricted administrative access.
Regional platform zone	Shared VPC host, regional subnets, KMS keys, service projects, perimeter and egress controls.	Residency-aligned project placement and private service access.
Domain production projects	Separate projects/datasets for data products, ingestion and processing according to scale and classification.	Least privilege, product ownership, quotas and cost allocation.
Non-production	Separate folders/projects, synthetic or masked data and lower quotas.	No uncontrolled production-sensitive data.
Recovery zone	Predefined recovery projects, configuration and data mechanisms according to tier.	Documented failover/failback and tested reconstruction.

7.4 Environment and Project Model

Environment	Purpose	Permitted data	Access	Lifecycle
Sandbox	Time-bound experimentation and learning.	Public, synthetic or explicitly approved masked data.	Individual/team groups; no production service identities.	Automatic expiry, quotas and no production dependencies.
Development	Build and unit/integration test.	Synthetic or minimized	Engineering groups and	Persistent while product is

Environment	Purpose	Permitted data	Access	Lifecycle
		representative data.	CI/CD identities.	active.
Integration/Test	Cross-service and contract validation.	Masked or approved controlled subsets.	Test teams, automation and limited engineers.	Aligned to release cycles.
Staging/Pre-production	Production-like performance, security and operational validation.	Masked or approved production-like data.	Restricted engineering, SRE and release identities.	Controlled and monitored.
Production	Authoritative governed services and products.	Approved business data by classification.	Consumer groups, production workloads and JIT administration.	Long-lived with formal change and recovery controls.
Recovery	Recovery of Tier 0/1 workloads.	Replicated or restorable approved data.	DR automation and emergency roles.	Warm/cold according to tier; tested regularly.

7.5 Primary End-to-End Data Flows

Flow 1 — Batch files

A source transfers a file and manifest through an approved private or controlled endpoint. The landing service verifies source identity, checksum, naming, malware policy and schema. Valid content is promoted to immutable raw storage; invalid content is quarantined with failure metadata. Orchestration loads or transforms data, applies quality and reconciliation, and publishes only after gates pass.

- Required evidence: source/consumer identity, data classification, owner, timestamps, release version and control outcome.
- Required failure behavior: retry boundary, dead-letter/quarantine path, reconciliation and runbook.
- Required operations: SLO, dashboard, alert owner, cost allocation and recovery procedure.

Flow 2 — Database CDC

Datastream captures supported database changes using least-privilege source accounts and private connectivity where available. Change events land in an approved target and are transformed into ordered, idempotent records. Periodic full reconciliation identifies missed, duplicated or out-of-order changes. Schema changes follow a contract process.

- Required evidence: source/consumer identity, data classification, owner, timestamps, release version and control outcome.
- Required failure behavior: retry boundary, dead-letter/quarantine path, reconciliation and runbook.
- Required operations: SLO, dashboard, alert owner, cost allocation and recovery procedure.

Flow 3 — Enterprise events

Producers publish versioned events to Pub/Sub. Dataflow validates, deduplicates, enriches and routes records. Invalid events move to a dead-letter path, and retained messages support replay. Event timestamps, processing timestamps and product timestamps support end-to-end freshness measurement.

- Required evidence: source/consumer identity, data classification, owner, timestamps, release version and control outcome.
- Required failure behavior: retry boundary, dead-letter/quarantine path, reconciliation and runbook.
- Required operations: SLO, dashboard, alert owner, cost allocation and recovery procedure.

Flow 4 — SQL product development

Engineers define transformations and assertions in Dataform. Pull requests trigger linting, dependency checks, unit/data tests and a compiled plan. Approved releases execute through controlled service identities and publish versioned tables or views. Lineage, ownership and release metadata are registered.

- Required evidence: source/consumer identity, data classification, owner, timestamps, release version and control outcome.
- Required failure behavior: retry boundary, dead-letter/quarantine path, reconciliation and runbook.
- Required operations: SLO, dashboard, alert owner, cost allocation and recovery procedure.

Flow 5 — Governed consumption

Consumers discover certified products in Knowledge Catalog and request access through the approved workflow. Authorization is granted to groups or workloads at the narrowest practical interface. Looker, BigQuery, sharing or APIs expose semantic or contractual views rather than unrestricted raw storage.

- Required evidence: source/consumer identity, data classification, owner, timestamps, release version and control outcome.
- Required failure behavior: retry boundary, dead-letter/quarantine path, reconciliation and runbook.
- Required operations: SLO, dashboard, alert owner, cost allocation and recovery procedure.

Flow 6 — AI and RAG

An approved AI workload consumes certified structured, document or feature products. Ingestion and chunking preserve document identity, classification and lineage. Embeddings and vector indexes use dedicated service identities. Retrieval, generation, safety, evaluation, citations and cost are logged according to risk tier.

- Required evidence: source/consumer identity, data classification, owner, timestamps, release version and control outcome.
- Required failure behavior: retry boundary, dead-letter/quarantine path, reconciliation and runbook.
- Required operations: SLO, dashboard, alert owner, cost allocation and recovery procedure.

7.6 Central Platform and Domain Responsibility Boundary

Capability	Central platform responsibility	Domain responsibility
Projects and network	Provide project factory, Shared VPC, private access and baseline policies.	Select approved pattern, supply ownership and request connectivity.
Ingestion	Provide connectors, templates, monitoring and failure framework.	Own source agreement, mapping, reconciliation and data-content issues.
Storage and processing	Provide zones, modules, quotas and service standards.	Design product schema, transformations and performance within standards.
Governance	Provide catalog, glossary model, classifications, workflows and conformance.	Maintain definitions, ownership, contracts, quality and lifecycle.
Security	Provide identity, KMS, perimeters, logs, guardrails and review.	Approve business access and maintain least-privilege consumer groups.
Analytics/AI	Provide governed delivery and AI platform patterns.	Own KPI/model/product logic, validation and consumer outcomes.
Operations	Provide telemetry platform, incident process and SRE standards.	Operate product SLO, runbook, quality and consumer communication.
FinOps	Provide billing data, allocation policy and optimization tooling.	Own product forecast, budget and remediation actions.

7.7 Strategic Google Cloud Service Profiles

The profiles below summarize the role of services that form the strategic reference architecture. Detailed quotas, regional availability, VPC Service Controls support and current commercial terms must be validated during implementation because service capabilities evolve.

7.7.1 BigQuery

Aspect	Architecture guidance
Purpose and role	Primary enterprise analytical warehouse, curated product store and SQL-serving layer.
Benefits	Serverless scale, separation of storage/compute, strong SQL ecosystem, fine-grained access, sharing and integration with Looker/Vertex AI.
Limitations	Poor table design and unrestricted scans can create cost/performance issues; procedural legacy logic may require redesign.

Aspect	Architecture guidance
Best practices, security and monitoring	Partition and cluster by access pattern; separate workloads; govern reservations; use row/column controls; monitor INFORMATION_SCHEMA and audit logs.
Alternatives and selection boundary	Self-managed warehouses, third-party cloud warehouses or Spark SQL; selected only where specific portability, engine or commercial requirements dominate.
Cost focus	Measure storage, processing, throughput, runtime, network transfer, logs or licensed users according to the service; allocate to product and owner.

7.7.2 Cloud Storage

Aspect	Architecture guidance
Purpose and role	Durable landing, raw, quarantine, archive and unstructured-data foundation.
Benefits	Global-scale object storage, lifecycle classes, integration and open-format support.
Limitations	Object semantics differ from file systems; many small objects and ungoverned buckets create cost and operational issues.
Best practices, security and monitoring	Use uniform bucket-level access, retention/lifecycle policy, naming/manifest standards, versioning where justified and private access.
Alternatives and selection boundary	BigQuery-native storage for analytical tables; specialized file systems only for workload-specific needs.
Cost focus	Measure storage, processing, throughput, runtime, network transfer, logs or licensed users according to the service; allocate to product and owner.

7.7.3 BigLake and managed Iceberg capabilities

Aspect	Architecture guidance
Purpose and role	Governed lakehouse access to open-format data and cross-engine scenarios.
Benefits	Centralized permissions and open-table interoperability while retaining data in object storage.
Limitations	Feature maturity, engine compatibility and transactional behavior must be validated for each workload.
Best practices, security and monitoring	Use when multi-engine or open-format value is explicit; test performance, metadata operations, security and recovery.
Alternatives and selection boundary	Native BigQuery tables for managed SQL analytics; direct bucket access is discouraged for broad consumption.
Cost focus	Measure storage, processing, throughput, runtime, network transfer, logs or licensed users according to the service; allocate to product and owner.

7.7.4 Pub/Sub

Aspect	Architecture guidance
Purpose and role	Enterprise event transport and decoupling layer for cloud-native streams.
Benefits	Managed scale, subscriptions, retention, replay and broad integration.
Limitations	Ordering, duplicates, schema evolution and downstream backpressure require explicit design.
Best practices, security and monitoring	Version events, use idempotent consumers, define retry/DLQ, monitor

Aspect	Architecture guidance
	backlog and retain messages according to recovery need.
Alternatives and selection boundary	Kafka may remain for existing ecosystem or protocol requirements; integration should be controlled.
Cost focus	Measure storage, processing, throughput, runtime, network transfer, logs or licensed users according to the service; allocate to product and owner.

7.7.5 Dataflow

Aspect	Architecture guidance
Purpose and role	Managed Apache Beam execution for streaming and complex parallel batch processing.
Benefits	Unified model, autoscaling, state/windowing and managed operations.
Limitations	Poorly designed keys, state or external calls can create hot keys, latency and cost.
Best practices, security and monitoring	Use Beam patterns, bounded state, dead-letter paths, templates, resource metrics and load testing.
Alternatives and selection boundary	BigQuery SQL/Dataform for ELT; Serverless Spark for existing Spark workloads.
Cost focus	Measure storage, processing, throughput, runtime, network transfer, logs or licensed users according to the service; allocate to product and owner.

7.7.6 Datastream

Aspect	Architecture guidance
Purpose and role	Serverless CDC from supported relational sources.
Benefits	Low-latency change capture with reduced operational burden.
Limitations	Supported source/target combinations, schema changes and source-log retention require validation.
Best practices, security and monitoring	Use least-privilege source accounts, private connectivity where supported, backfill plan and periodic reconciliation.
Alternatives and selection boundary	Debezium/Kafka or commercial CDC where source support or transformation requirements demand it.
Cost focus	Measure storage, processing, throughput, runtime, network transfer, logs or licensed users according to the service; allocate to product and owner.

7.7.7 Dataform

Aspect	Architecture guidance
Purpose and role	Versioned BigQuery SQL transformation, dependency and assertion framework.
Benefits	Git-based development, compilable dependency graph and warehouse-centric workflow.
Limitations	Not a general-purpose cross-service orchestrator or streaming engine.
Best practices, security and monitoring	Use standardized repositories, assertions, release configuration, service identities and CI validation.
Alternatives and selection boundary	dbt or orchestrator SQL tasks where enterprise tooling standards dictate.
Cost focus	Measure storage, processing, throughput, runtime, network transfer, logs or licensed users according to the service; allocate to product and owner.

7.7.8 Managed Service for Apache Airflow

Aspect	Architecture guidance
Purpose and role	Complex schedule and dependency orchestration across data services.
Benefits	Managed Airflow ecosystem and DAG model.
Limitations	Environment sizing, dependency management, DAG quality and

Aspect	Architecture guidance
	scheduler health require disciplined operations.
Best practices, security and monitoring	Use only for orchestration; keep transformations in target engines; enforce DAG standards and environment isolation.
Alternatives and selection boundary	Workflows for simpler service orchestration; Dataform for SQL dependency execution.
Cost focus	Measure storage, processing, throughput, runtime, network transfer, logs or licensed users according to the service; allocate to product and owner.

7.7.9 Knowledge Catalog

Aspect	Architecture guidance
Purpose and role	Enterprise metadata, discovery, glossary, lineage and data-product registration layer.
Benefits	Native asset discovery and governance integration across Google Cloud.
Limitations	External assets and custom lineage may require integration; metadata quality still depends on ownership.
Best practices, security and monitoring	Automate registration, define aspect types, certify products and measure completeness.
Alternatives and selection boundary	Third-party catalog where cross-cloud breadth or established enterprise investment outweighs native integration.
Cost focus	Measure storage, processing, throughput, runtime, network transfer, logs or licensed users according to the service; allocate to product and owner.

7.7.10 Sensitive Data Protection

Aspect	Architecture guidance
Purpose and role	Discovery, inspection and transformation support for sensitive information.
Benefits	Built-in detectors and integration with cloud data services.
Limitations	False positives/negatives require tuning; inspection does not itself enforce every access policy.
Best practices, security and monitoring	Use sampling strategy, custom infoTypes, classification workflow and human review for critical classes.
Alternatives and selection boundary	Third-party data-security platforms or custom detection for specialized patterns.
Cost focus	Measure storage, processing, throughput, runtime, network transfer, logs or licensed users according to the service; allocate to product and owner.

7.7.11 Vertex AI

Aspect	Architecture guidance
Purpose and role	Managed platform for model development, deployment, pipelines, registry, evaluation and generative AI.
Benefits	Integrated ML lifecycle and access to Google models/services.
Limitations	Model, region, networking, quota, safety and cost constraints vary by capability.
Best practices, security and monitoring	Use dedicated projects/identities, approved data products, evaluations, model cards, endpoint controls and cost monitoring.
Alternatives and selection boundary	Specialized external ML platforms where model ecosystem, sovereignty or portability requires.
Cost focus	Measure storage, processing, throughput, runtime, network transfer, logs or licensed users according to the service; allocate to product and owner.

7.7.12 Looker

Aspect	Architecture guidance
Purpose and role	Governed enterprise semantic layer, metrics and analytical application platform.

Aspect	Architecture guidance
Benefits	Reusable business definitions, access controls and governed exploration.
Limitations	Poor LookML design, uncontrolled content and query patterns can reduce performance and trust.
Best practices, security and monitoring	Central metric governance, content lifecycle, testing, caching strategy and product-aligned models.
Alternatives and selection boundary	Looker Studio for lightweight use; other BI products during controlled coexistence.
Cost focus	Measure storage, processing, throughput, runtime, network transfer, logs or licensed users according to the service; allocate to product and owner.

7.8 Reference Architecture Conformance

- A solution may omit services that are not required; it may not add services without a clear capability and operating owner.
- A solution must preserve the control intent even when an alternative service is approved.
- Every implementation supplies a data-flow diagram, trust boundaries, region placement, IAM model, SLOs, failure modes and cost allocation.
- Material deviations are documented through an ADR and time-bound exception where necessary.
- Conformance is reassessed after material scale, classification, consumer, region or AI-use changes.

8. Architecture Decision Records

Architecture Decision Records capture the durable decisions that shape the platform. Each decision states the problem, alternatives, trade-offs, risks, mitigation and review triggers so future teams can understand both the choice and its boundaries.

Section outcomes

Twenty baseline decisions; Explicit alternatives; Trade-offs and risks; Review triggers.

8.1 ADR-001 — Adopt a product-based enterprise data platform

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	Fragmented business-unit platforms duplicate capabilities, controls and operating effort while failing to provide a consistent consumer experience.
Alternatives considered	A) Fully centralized delivery team; B) Independent domain platforms; C) Central self-service platform with domain-owned products.
Decision	Adopt a central platform product that supplies paved-road capabilities, combined with federated domain ownership of data meaning, quality, contracts and product SLOs.
Rationale	The model preserves enterprise control and reuse without making the central team a permanent delivery bottleneck.
Trade-offs	Requires mature product management, delegated accountability and automated governance. Domains must fund product ownership rather than treating data as an engineering by-product.
Key risks	Central platform may over-standardize; domains may ignore ownership; capability overlap may persist.
Mitigations	Publish responsibility boundaries, measure reuse, create governance RACI, fund platform backlog and enforce exceptions through architecture review.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review if domain adoption stalls, central backlog exceeds agreed lead time, or major regulatory requirements require centralized ownership.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.2 ADR-002 — Use BigQuery as the primary analytical warehouse

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	The enterprise requires petabyte-scale SQL analytics, workload isolation, governed sharing and reduced infrastructure administration.
Alternatives considered	A) Expand legacy warehouse; B) Self-managed Spark/Hadoop SQL; C) Third-party cloud warehouse; D) BigQuery.
Decision	Use BigQuery as the default curated analytical store and SQL-serving layer. Alternatives require workload-specific justification.
Rationale	BigQuery aligns with serverless-first, separation of storage and compute, integrated governance, Looker and Vertex AI, and elastic global operations.
Trade-offs	Teams must learn BigQuery optimization and economics; some procedural or engine-specific workloads require redesign; portability is not cost-free.
Key risks	Uncontrolled scans and concurrency increase cost; poor partition design reduces performance; legacy migration complexity is underestimated.
Mitigations	SQL standards, partition and clustering review, reservations where justified, workload projects, cost controls, benchmarks and migration assessment.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review for material engine gaps, sovereignty constraints, commercial change or sustained workload characteristics that favor another engine.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.3 ADR-003 — Use Cloud Storage and BigLake for open-format lakehouse needs

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	Some workloads require immutable raw retention, unstructured content, archival and cross-engine access that a warehouse-only model does not address.
Alternatives considered	A) BigQuery-only; B) Direct bucket access; C) Cloud Storage with BigLake governance; D) Self-managed lakehouse platform.
Decision	Use Cloud Storage for landing, raw and archive and BigLake-managed access for approved open-format or multi-engine workloads; use native BigQuery tables for managed analytical serving by default.
Rationale	This balances open formats and storage economics with centralized access and governance.
Trade-offs	Two storage paradigms increase design choices; open-table compatibility and metadata behavior require testing.
Key risks	An uncontrolled data lake, duplicate truth, small-file problems and inconsistent security.
Mitigations	Layer standards, product ownership, format decision criteria, compaction, managed access, catalog registration and lifecycle policy.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review as managed Iceberg and BigLake capabilities evolve or when cross-engine demand changes materially.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.4 ADR-004 — Use regional residency zones with controlled global discovery

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	A single global data location cannot satisfy all sovereignty, latency and recovery requirements.
Alternatives considered	A) One global multi-region; B) Independent regional platforms; C) Regional data zones with common global controls.
Decision	Deploy data and processing in approved regional zones, with consistent organization policy, metadata standards and controlled product exchange.
Rationale	The design supports residency and regional operations while avoiding divergent architectures.
Trade-offs	Cross-region product composition is more complex; duplicate reference data and recovery copies may be necessary.
Key risks	Unauthorized transfers, inconsistent regional service availability and unclear ownership.
Mitigations	Residency classification, region policy, transfer register, regional service validation, legal approval and monitored egress.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review when laws, corporate structure, service regions or business-continuity targets change.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.5 ADR-005 — Use Shared VPC and private service access patterns

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	Domain projects require consistent network control without central teams managing every workload resource.
Alternatives considered	A) Standalone VPC per project; B) Flat shared project; C) Shared VPC host with service projects and Private Service Connect.
Decision	Use regional Shared VPC host projects, attached service projects, hierarchical firewall policies, Private Google Access and Private Service Connect where supported.
Rationale	Central networking control is separated from workload ownership and supports consistent connectivity, inspection and address management.
Trade-offs	Coordination between network and application teams is required; quotas and attachment models must be planned.
Key risks	The central network team becomes a bottleneck; transitive paths or DNS errors expose data or block services.
Mitigations	Automated network modules, published connectivity products, flow logs, DNS standards and pre-approved patterns.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review for major network topology, acquisition or multi-cloud integration changes.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.6 ADR-006 — Use VPC Service Controls for restricted data perimeters

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	IAM alone does not sufficiently reduce data-exfiltration risk from supported managed services.
Alternatives considered	A) IAM only; B) Network controls only; C) VPC Service Controls plus IAM and egress governance.
Decision	Place restricted-data projects and supported services in designed VPC Service Controls perimeters, using access levels, ingress and egress rules and staged rollout.
Rationale	Service perimeters add a control against unintended movement of managed-service data outside approved boundaries.
Trade-offs	Integrations become more complex, and unsupported interfaces require alternative controls.
Key risks	Production disruption, overly permissive egress, and false confidence where services are unsupported.
Mitigations	Dry-run mode, interface inventory, automated tests, explicit egress rules, break-glass procedure and current support validation.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review when services, integrations or classification change.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.7 ADR-007 — Apply classification-based CMEK rather than universal CMEK

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	Key-control requirements differ by data sensitivity, regulation and customer commitments.
Alternatives considered	A) Google-managed encryption only; B) CMEK everywhere; C) Selective CMEK by classification and workload.
Decision	Use Google-managed encryption for standard workloads and CMEK for restricted, regulated, contractually controlled or high-impact assets.
Rationale	Selective CMEK provides required control without making key availability and administration a universal dependency.
Trade-offs	Classification must be accurate; mixed encryption patterns require policy automation and clear ownership.
Key risks	Key deletion or permission errors cause outage; inconsistent adoption; key-region mismatch.
Mitigations	Central key hierarchy, separation of duties, rotation, service-agent tests, deletion protection, monitoring and recovery runbook.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review for regulatory, contractual or product changes and new service CMEK support.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.8 ADR-008 — Use Pub/Sub as the default cloud-native event backbone

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	Producers and consumers require scalable asynchronous integration, replay and independent change.
Alternatives considered	A) Direct service calls; B) Self-managed Kafka; C) Managed Kafka; D) Pub/Sub.
Decision	Use Pub/Sub for new Google Cloud-native event streams; retain or integrate Kafka where ecosystem, protocol or migration requirements justify it.
Rationale	Pub/Sub reduces infrastructure operations and integrates with Dataflow and Google Cloud services.
Trade-offs	Kafka semantics and tooling do not map one-to-one; application teams must design idempotency and ordering consciously.
Key risks	Schema drift, duplicates, backlog growth and retention insufficient for recovery.
Mitigations	Event contracts, schema and version governance, dead-letter topics, replay tests, backlog alerts and capacity tests.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review when protocol, transactional or ecosystem requirements cannot be met.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.9 ADR-009 — Use Dataflow for complex streaming and parallel data processing

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	The enterprise needs stateful event-time processing, autoscaling and common engineering patterns for streams and complex batch.
Alternatives considered	A) Custom consumers; B) Spark streaming; C) SQL-only processing; D) Dataflow and Apache Beam.
Decision	Use Dataflow for stateful, high-throughput or complex streaming and for Beam-based parallel batch; use simpler direct delivery where transformation is minimal.
Rationale	Managed Beam execution supports reusable code, windowing, state and operational scale.
Trade-offs	Beam expertise is required; poor key and state design can be costly; small simple workloads may be over-engineered.
Key risks	Hot keys, external-call bottlenecks, duplicate side effects and opaque cost.
Mitigations	Template library, coding standards, load tests, bounded state, asynchronous I/O patterns, metrics and cost allocation.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review when SQL, Spark or direct subscription patterns meet the workload more simply.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.10 ADR-010 — Use Dataform for governed BigQuery ELT

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	SQL transformations need dependencies, version control, testing and repeatable release management.
Alternatives considered	A) Ad hoc SQL; B) Airflow SQL tasks; C) dbt; D) Dataform.
Decision	Use Dataform as the default BigQuery SQL transformation framework, subject to enterprise tooling and feature validation.
Rationale	It aligns transformations closely with BigQuery, supports versioned development and assertions, and avoids using orchestration as a transformation engine.
Trade-offs	Teams with existing dbt estates may require coexistence; cross-service orchestration remains separate.
Key risks	Repository sprawl, weak testing and direct production changes.
Mitigations	Repository templates, release configuration, CI compilation, assertions, protected branches and service-account boundaries.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review if enterprise standardization on another compatible transformation framework creates greater value.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.11 ADR-011 — Use Managed Service for Apache Airflow selectively for complex orchestration

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	Cross-service pipelines need scheduling, dependency management, backfill and operational visibility.
Alternatives considered	A) Custom scheduler; B) Workflows; C) Dataform scheduling; D) Managed Service for Apache Airflow.
Decision	Use Managed Service for Apache Airflow for complex DAGs; use Workflows or native scheduling for simpler sequences and Dataform for SQL dependencies.
Rationale	A tiered orchestration approach prevents Airflow from becoming a universal execution engine.
Trade-offs	Multiple orchestration mechanisms require clear decision criteria and shared observability.
Key risks	DAG bloat, embedded business logic, dependency conflicts and oversized environments.
Mitigations	Thin-DAG standard, environment separation, dependency controls, DAG testing and workload-placement review.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review based on DAG portfolio complexity, service evolution and operating cost.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.12 ADR-012 — Use Knowledge Catalog as the native governance and discovery layer

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	Users need searchable assets, common terminology, ownership, lineage and certification across platform products.
Alternatives considered	A) Wiki or spreadsheet; B) Third-party catalog; C) Knowledge Catalog; D) Hybrid catalog.
Decision	Use Knowledge Catalog for native Google Cloud asset governance and data-product registration, integrating enterprise catalog tooling when cross-platform scope requires it.
Rationale	Native harvesting and integration reduce duplication for Google Cloud assets.
Trade-offs	External estates and custom metadata may require integration; a tool cannot replace stewardship.
Key risks	Incomplete metadata, duplicate glossaries and a catalog with low adoption.
Mitigations	Mandatory product metadata, automated completeness checks, glossary governance, owner workflow and consumer-experience measures.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review if cross-cloud catalog requirements or strategic enterprise tooling change.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.13 ADR-013 — Adopt federated data mesh principles

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	A single central team cannot sustainably own business meaning and quality across all enterprise domains.
Alternatives considered	A) Central data team; B) Fully decentralized teams; C) Federated data mesh with self-service platform.
Decision	Adopt domain-oriented data ownership, data as product, self-service platform and federated computational governance.
Rationale	The model places accountability near business knowledge while retaining enterprise interoperability and control.
Trade-offs	Requires domain funding, product management and shared standards; organizational change is substantial.
Key risks	Data mesh becomes uncontrolled project sprawl or a relabeling exercise without ownership.
Mitigations	Domain criteria, product contract, platform boundary, governance RACI, maturity gates and measurable adoption.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review if business structure or accountability model changes.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.14 ADR-014 — Use Looker as the strategic governed semantic layer

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	Different BI tools and reports implement inconsistent KPI definitions and access logic.
Alternatives considered	A) Definitions in each dashboard; B) BigQuery views only; C) Looker semantic model; D) Third-party semantic layer.
Decision	Use Looker for enterprise governed metrics and semantic modeling while allowing approved consumption tools to access certified interfaces.
Rationale	Central reusable semantics improve consistency, security and analytical productivity.
Trade-offs	LookML governance and model ownership are required; not every lightweight report needs a complex model.
Key risks	Monolithic models, dashboard sprawl and query inefficiency.
Mitigations	Domain-aligned models, KPI council, content lifecycle, testing, performance review and certified content.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review for enterprise BI strategy or semantic-layer capability changes.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.15 ADR-015 — Use Vertex AI as the strategic managed AI platform

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	AI teams need governed experimentation, pipelines, registry, deployment, generative AI and monitoring without building separate infrastructure.
Alternatives considered	A) Self-managed ML stack; B) Third-party platform; C) Vertex AI; D) Workload-specific mixed platforms.
Decision	Use Vertex AI for new governed machine-learning and generative-AI workloads; approve alternatives for model, sovereignty or ecosystem gaps.
Rationale	It integrates managed ML lifecycle capabilities with Google Cloud data and security services.
Trade-offs	Capability availability, region, model terms and cost require workload-level review.
Key risks	Sensitive-data leakage, untested prompts or models, runaway token cost and vendor dependence.
Mitigations	AI risk tiers, approved data products, dedicated identities, evaluations, guardrails, model cards, monitoring and budgets.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review per major model release, risk-class change or service-capability change.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.16 ADR-016 — Use Terraform and policy-as-code for platform provisioning

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	Manual configuration cannot deliver repeatable, reviewable and recoverable enterprise foundations.
Alternatives considered	A) Console and manual; B) legacy deployment tools; C) Terraform; D) custom API automation.
Decision	Use Terraform modules, versioned state and policy-as-code as the standard for organization, projects and platform resources.
Rationale	Terraform supports reusable modules, reviewable plans and broad ecosystem adoption.
Trade-offs	State security, provider upgrades and module lifecycle require disciplined ownership.
Key risks	State compromise, destructive plan, drift and module coupling.
Mitigations	Protected remote state, separation of plan and apply, approvals, deletion protection, tests, version pinning and drift scans.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review if alternative tooling materially improves enterprise outcomes.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.17 ADR-017 — Use classification and service tier as primary control dimensions

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	Applying the strongest controls to every asset is costly, while weak uniform controls expose critical data.
Alternatives considered	A) One control baseline; B) Team-defined controls; C) Classification multiplied by service-tier policy matrix.
Decision	Assign every product a data classification and service tier; derive access, encryption, residency, logging, SLO and recovery requirements from the matrix.
Rationale	A risk-based model creates proportional, explainable controls.
Trade-offs	Misclassification creates risk; the matrix must be maintained and automated.
Key risks	Teams choose lower tiers to reduce cost; inherited classifications are lost through transformation.
Mitigations	Owner approval, automated propagation, governance sampling and escalation for inconsistent selections.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review annually or after regulation or risk-model changes.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.18 ADR-018 — Manage critical data products with SLOs and error budgets

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	Component uptime does not express consumer experience, and teams lack a rational balance between reliability and change.
Alternatives considered	A) Provider SLA only; B) ticket-based operations; C) product SLOs and error budgets.
Decision	Define availability, freshness, quality and latency SLIs and SLOs for Tier 0 and Tier 1 data products and use error budgets in release and remediation decisions.
Rationale	This focuses operations on user outcomes and makes reliability trade-offs explicit.
Trade-offs	Instrumentation and ownership require investment; product SLOs may expose dependencies outside platform control.
Key risks	Vanity SLOs, incomplete measurement and ignored budget exhaustion.
Mitigations	Standard SLO catalog, review cadence, dependency objectives, automated reporting and governance escalation.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review SLOs quarterly and after significant consumer or architecture changes.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.19 ADR-019 — Use showback before chargeback

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	The enterprise needs cost accountability, but early allocation rules and unit economics may be unstable.
Alternatives considered	A) Central absorption; B) immediate chargeback; C) showback followed by policy-based chargeback.
Decision	Implement detailed showback first; introduce chargeback only after metadata coverage, allocation accuracy and governance are proven.
Rationale	Teams gain visibility and behavior signals without disputes undermining initial adoption.
Trade-offs	Central funding continues temporarily; cost incentives may be weaker before chargeback.
Key risks	Showback never matures; shared costs remain disputed.
Mitigations	Set readiness criteria, allocation KPIs, executive sponsor and a staged chargeback roadmap.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review quarterly until chargeback readiness is achieved.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

8.20 ADR-020 — Use tiered disaster-recovery patterns

Decision element	Record
Status	Proposed enterprise baseline; becomes accepted after Architecture Review Board approval.
Architecture domain	Enterprise data platform
Problem	A single disaster-recovery pattern is either too costly for low-impact workloads or insufficient for critical workloads.
Alternatives considered	A) Uniform active-active; B) backup-only; C) tiered recovery based on impact.
Decision	Apply Tier 0 to Tier 3 recovery patterns with explicit RTO and RPO, service-specific mechanisms, infrastructure reconstruction and scheduled tests.
Rationale	The approach aligns recovery investment with business impact and service capability.
Trade-offs	Portfolio governance and dependency mapping are more complex; not every managed service fails over identically.
Key risks	Paper compliance, hidden dependencies and untested regional assumptions.
Mitigations	Recovery architecture per product, exercise calendar, evidence, dependency tests and executive acceptance of residual risk.
Implementation implications	Represent the decision in standards, reusable modules, design reviews, tests, operational procedures and the service catalog.
Cost implications	Include both consumption cost and operational effort in business cases and product forecasts.
Security implications	Assess threat model, identity, classification, network path, logging and evidence before production.
Review trigger	Review annually, after recovery tests, incidents or service-topology changes.
Review cadence	At least annually and after material incident, regulation change, service retirement or strategy change.

Acceptance criterion: a representative implementation demonstrates this decision under normal operation, expected peak load, a defined failure scenario, and the applicable security and operational controls.

9. Technology Selection Matrix

The technology matrix converts architecture principles into practical defaults. A default is not a mandatory product for every workload; it is the starting point that reduces repeated evaluation. Alternatives require evidence that they better satisfy documented requirements and that their operating model is funded.

Section outcomes

Strategic defaults; Selection boundaries; Alternatives; Exception criteria.

Capability	Default technology	Selection rationale	Alternatives	Exception criterion	Mandatory design concerns
Analytical warehouse	BigQuery	Petabyte-scale governed SQL analytics, curated products and BI serving.	Existing warehouse; third-party cloud warehouse; Spark SQL.	Choose an alternative for unsupported features, sovereignty, contractual or proven economics/performance requirements.	Partitioning, reservations, IAM, data quality, cost attribution.
Raw object storage	Cloud Storage	Landing, immutable raw, quarantine, archive and documents.	BigQuery native storage; on-premises object store.	Alternative only where source locality, protocol or legal requirement prevents use.	Uniform access, retention, lifecycle, naming, event controls.
Open lakehouse	BigLake with supported open-table capabilities	Cross-engine, open-format and managed-access use cases.	Direct buckets; self-managed Iceberg; native BigQuery.	Use only when open/multi-engine value exceeds complexity.	Catalog, compaction, metadata, access and recovery testing.
Event transport	Pub/Sub	New cloud-native asynchronous events and decoupled streams.	Kafka; direct API; managed Kafka.	Kafka for ecosystem/protocol requirements or controlled migration.	Schemas, idempotency, retention, DLQ, backlog SLO.
CDC	Datastream	Supported relational change capture with low operational burden.	Debezium/Kafka; commercial CDC; batch extract.	Alternative for unsupported sources, transformations or enterprise CDC standard.	Source-log retention, private connectivity, backfill and reconciliation.
Streaming processing	Dataflow and Apache Beam	Stateful event-time processing, enrichment and complex streams.	Spark streaming; custom services; SQL.	Alternative for existing Spark skill/estate or simple direct delivery.	Hot-key tests, state bounds, replay, templates and cost.
Parallel batch	Dataflow or Serverless for Apache Spark	Complex parsing or distributed processing.	BigQuery SQL; Dataproc cluster; custom compute.	Use SQL for warehouse-native transformation; Spark for established libraries.	Autoscaling, ephemeral compute, output idempotency.
BigQuery ELT	Dataform	Versioned SQL dependencies, assertions and releases.	dbt; Airflow SQL; scripts.	Alternative if enterprise standard/tooling has stronger lifecycle integration.	Repository standard, CI compile, assertions and service identity.
Complex orchestration	Managed Service for Apache Airflow	Cross-service DAGs, schedules and backfills.	Workflows; native schedulers; Dataform.	Use simpler mechanisms where workflow complexity does not justify Airflow.	Thin DAGs, isolated environments, dependency management.
Simple orchestration	Workflows	Service/API coordination with limited state and scheduling.	Cloud Run; Airflow; custom code.	Use Airflow for complex data DAGs; Cloud Run for custom service logic.	Retry policy, identity, timeout and error path.
Metadata and discovery	Knowledge Catalog	Native catalog, glossary, lineage and product registration.	Third-party catalog; hybrid.	Hybrid for enterprise-wide cross-cloud catalog requirements.	Mandatory metadata, aspect standards and completeness checks.
Sensitive-data discovery	Sensitive Data Protection	Identification and transformation support for sensitive data.	Third-party DSPM/DLP; custom detectors.	Alternative for specialized detectors or consolidated security platform.	Custom infoTypes, sampling, review and policy integration.
Enterprise semantic layer	Looker	Governed measures, dimensions and analytical applications.	BigQuery views; third-party semantic/BI platform.	Coexist during migration or for approved specialized capabilities.	Model ownership, testing, content lifecycle and performance.
Lightweight visualization	Looker Studio	Low-complexity dashboards and	Looker; Connected Sheets; third-party BI.	Not the source of enterprise KPI logic.	Certified sources, sharing controls and owner.

Capability	Default technology	Selection rationale	Alternatives	Exception criterion	Mandatory design concerns
		collaboration.			
Spreadsheet analysis	Connected Sheets	Governed spreadsheet access to BigQuery.	Extracted files; local spreadsheets.	File export only under approved data-sharing policy.	Data source certification, access inheritance, query limits.
AI/ML platform	Vertex AI	Managed training, pipelines, registry, endpoints and generative AI.	Self-managed ML; third-party platform.	Alternative for model ecosystem, sovereignty or commercial requirements.	Risk tier, evaluations, identity, network, monitoring and budget.
Vector retrieval	Vertex AI Vector Search or validated BigQuery vector capabilities	Enterprise RAG and similarity search.	Open-source vector DB; specialist managed service.	Select using scale, latency, filtering, region and integration tests.	Index lifecycle, embedding version, evaluation and access.
API runtime	Cloud Run	Stateless data APIs, integration adapters and control services.	GKE; Functions; Apigee runtime.	GKE for platform/runtime requirements; Functions for event handlers.	Private ingress, service identity, concurrency, timeout and tracing.
API management	Apigee or API Gateway by complexity	Protected internal/external data APIs and policy enforcement.	Cloud Run direct; third-party gateway.	Apigee for enterprise lifecycle/monetization; API Gateway for simpler APIs.	Authentication, quotas, threat protection, audit and developer lifecycle.
Infrastructure as Code	Terraform	Organization, landing zone, platform and domain resources.	Manual/API automation; alternative IaC.	Exception for unsupported resource only, with reconciliation plan.	Remote state, plan/apply separation, tests and version pinning.
CI/CD	Cloud Build and/or GitHub Actions	Build, test, security scan and promotion.	Jenkins; GitLab; other enterprise tooling.	Use enterprise standard where identity/provenance/control requirements are met.	OIDC/WIF, immutable artifacts, approvals and audit.
Artifact management	Artifact Registry	Containers and build artifacts.	External registry; source releases.	Alternative for global enterprise standard or regulated separation.	Vulnerability scan, retention, immutability and least privilege.
Monitoring and logging	Cloud Monitoring and Cloud Logging	Native telemetry, dashboards, alerts and audit integration.	Enterprise observability suite.	Export/integrate where enterprise operations standard requires.	SLOs, log routing, retention, exclusions and cost.
Security posture	Security Command Center	Cloud findings, posture and threat signals.	Enterprise CNAPP/SIEM; hybrid.	Integrate rather than duplicate when enterprise SOC has a strategic platform.	Organization scope, triage ownership, exports and response.

9.1 Selection Method

Criterion	Weight range	Required evidence
Functional fit	20–30%	Traceable requirements, supported interfaces and feature validation.
Security and compliance	15–25%	Control mapping, identity/network model, data handling and evidence.
Reliability and recovery	10–20%	Failure modes, dependency behavior, RTO/RPO and test plan.
Scale and performance	10–20%	Representative benchmark, quota model and growth capacity.
Operating model	10–15%	Ownership, skill, support, automation and incident procedures.
Economics	10–20%	Consumption model, unit economics, migration and operating effort.
Portability and lifecycle	5–10%	Data/code export, standards, product maturity and exit plan.

Weights are adjusted by workload. For a regulated Tier 0 product, security and recovery receive greater weight; for a disposable experiment, speed and cost may dominate. A technology decision is valid only within the workload assumptions tested.

9.2 Exception Governance

- State the unmet requirement or material advantage that the default cannot satisfy.
- Compare the alternative using the same criteria and realistic operating effort.
- Identify owner, funding, skills, security controls, monitoring, recovery and exit plan.
- Set an expiry or decision-review date; permanent exceptions are incorporated into the strategic catalog.
- Record migration impact and interoperability with data products, identity, networking and governance.

10. Google Cloud Service Mapping

Service mapping shows how business and platform capabilities are realized by strategic Google Cloud services. It prevents tool-centric designs by making the supported capability and required controls explicit.

Section outcomes

Capability-to-service traceability; Service roles; Control overlays; Operating ownership.

Capability	Primary services	Architecture role	Mandatory controls	Operating owner
Source transfer	Storage Transfer Service, Transfer Appliance where justified	Bulk object transfer, scheduled movement and initial migration.	Private/approved endpoints, checksum, manifest, residency and reconciliation.	Ingestion platform team
Relational CDC	Datastream	Supported database change capture and backfill.	Least-privilege source account, source-log retention, private path, schema handling.	Ingestion platform and source owner
Events	Pub/Sub	Topics, subscriptions, decoupling, retention and replay.	IAM, schema/version, DLQ, backlog, encryption and perimeter support.	Event platform team
Stream/batch transformation	Dataflow	Beam pipelines, state/windowing, enrichment and distributed batch.	Service identity, network, template provenance, metrics, replay and cost.	Data engineering platform
SQL transformations	Dataform and BigQuery	Versioned dependency graph, assertions and warehouse ELT.	Protected repositories, release configuration, service identity and audit.	Analytics engineering
Spark processing	Serverless for Apache Spark / Dataproc	Existing Spark libraries, complex data science and distributed jobs.	Ephemeral compute, network, image/dependency control and cost.	Data engineering platform
Orchestration	Managed Service for Apache Airflow	Complex data DAGs and backfills.	Environment isolation, DAG testing, secret handling, scheduler health.	Orchestration platform
Service workflows	Workflows	API/service coordination and simple state machines.	Least-privilege identity, retry, timeout and traceability.	Platform/application team
Raw and archive	Cloud Storage	Landing, quarantine, immutable raw, documents and archive.	Uniform access, retention, lifecycle, CMEK policy, logging.	Domain with platform controls
Analytical warehouse	BigQuery	Standardized/curated tables, views, SQL serving and sharing.	Dataset IAM, policy tags, reservations, audit, recovery and cost.	Domain data-product team
Open lakehouse	BigLake	Managed access to open-format data.	Metadata, table format, access, compaction and engine validation.	Platform and domain
Metadata and discovery	Knowledge Catalog	Catalog, glossary, ownership, lineage and product metadata.	Aspect standards, completeness, stewardship and lifecycle.	Governance platform
Sensitive-data control	Sensitive Data Protection	Inspection, classification support and de-identification.	Detector tuning, sampling, review and policy integration.	Security and privacy
Semantic analytics	Looker	LookML models, governed metrics and analytical applications.	Content governance, access, caching, tests and usage.	Analytics product team
Light BI	Looker Studio	Lightweight reports and collaboration.	Certified data, sharing and lifecycle controls.	Domain analytics
Spreadsheet analytics	Connected Sheets	Governed BigQuery access from spreadsheets.	Access inheritance, source certification and query controls.	Business analytics
Model lifecycle	Vertex AI	Workbench, training, pipelines, registry, endpoints and monitoring.	AI risk tier, identity, network, evaluations and audit.	AI platform and model owner
Generative AI	Vertex AI Gemini capabilities	Grounded generation, evaluation and enterprise applications.	Prompt/data controls, safety, citations, red-team and token budget.	AI platform and use-case owner

Capability	Primary services	Architecture role	Mandatory controls	Operating owner
RAG retrieval	Vector Search / BigQuery vector capabilities	Embedding indexing and similarity retrieval.	Embedding/model version, index ACL, freshness and evaluation.	AI platform
Data/API services	Cloud Run	Data APIs, adapters and control services.	Private ingress, service identity, secrets, tracing and autoscaling.	Application/data platform
API governance	Apigee / API Gateway	Authentication, policy, quotas and lifecycle for data APIs.	Threat protection, developer access, audit and availability.	Integration platform
Identity	Cloud IAM, Cloud Identity, Workforce/Workload Identity Federation	Human and workload authentication/authorization.	Groups, JIT, impersonation, no long-lived keys, access reviews.	Identity/security
Network	Shared VPC, PSC, Interconnect, VPN, Cloud DNS, firewall policies	Private connectivity and segmentation.	Address plan, routing, DNS, firewall hierarchy, flow logs.	Network platform
Keys and secrets	Cloud KMS, Cloud HSM, Secret Manager	Key lifecycle, CMEK and application secrets.	Separation of duties, rotation, access logging and recovery.	Security platform
Security posture	Security Command Center	Findings, threat detection and cloud posture.	Organization enablement, triage, SIEM export and remediation.	Security operations
Observability	Cloud Monitoring, Cloud Logging, Trace and Error Reporting	Metrics, logs, traces, alerts and SLOs.	Log routing, retention, alert ownership and cost.	SRE/platform
Delivery automation	Terraform, Cloud Build/GitHub Actions, Artifact Registry, Cloud Deploy where relevant	Infrastructure and application delivery.	WIF, provenance, policy, immutable artifacts, approvals and rollback.	DevSecOps
FinOps	Cloud Billing export, BigQuery, dashboards and reservations	Allocation, forecast, optimization and unit economics.	Labels/tags, ownership, data quality and review cadence.	FinOps

10.1 Service-use Rules

- Service enablement occurs through approved project and Terraform modules, not ad hoc console use.
- Each service instance or logical deployment has an owner, environment, classification, cost center and support tier.
- Private access, service perimeter, CMEK and region behavior are validated for the exact service capability used.
- Audit, operational and cost telemetry are routed to enterprise platforms before production acceptance.
- Services that store or process business data are registered in the architecture repository and data catalog as applicable.
- Unsupported or preview capabilities require explicit risk acceptance and cannot silently underpin Tier 0 commitments.

11. Enterprise Capability Matrix

The capability matrix defines what the platform must enable and the maturity expected over the transformation. It is used for investment planning, architecture assurance and quarterly maturity reviews.

Section outcomes

Capability scope; Current-to-target maturity; Accountability; Evidence.

Domain	Capability	Current maturity	Target maturity	Target-state evidence	Accountable owner
Strategy and governance	Platform product management	1	4	Funded roadmap, adoption telemetry and quarterly value review.	Platform Product Owner
Strategy and governance	Architecture governance	2	4	Strategic service catalog, ADRs, exception lifecycle and conformance reporting.	Chief Architect
Strategy and governance	Data governance council	2	4	Decision rights, glossary, classification and domain accountability.	CDO
Foundation	Organization and policy hierarchy	2	5	Automated folder/project deployment and policy assurance.	Cloud Platform
Foundation	Project factory	1	5	Self-service governed projects delivered within two working days.	Cloud Platform
Foundation	Hybrid connectivity	2	4	Resilient private connectivity, routing, DNS and monitoring.	Network Platform
Foundation	Identity and privileged access	2	5	Federation, JIT privilege, WIF and key-elimination evidence.	Security/Identity
Foundation	Service perimeter and egress governance	1	4	Dry-run validation, controlled rules and exfiltration monitoring.	Security/Network
Data acquisition	Batch ingestion	2	5	Reusable manifest, validation, quarantine and reconciliation pattern.	Data Platform
Data acquisition	CDC	1	4	Supported source factory with schema handling and replay/reconciliation.	Data Platform
Data acquisition	Streaming events	2	4	Contracted events, DLQ, replay, backlog SLO and templates.	Event Platform
Data acquisition	API/SaaS integration	2	4	Approved adapter patterns, secrets, retries and source telemetry.	Integration Platform
Data foundation	Raw and archive management	2	5	Immutable raw standards, lifecycle, legal hold and recovery.	Data Platform
Data foundation	BigQuery analytical products	2	5	Domain datasets, performance standards, access and recovery.	Domain/Product Teams
Data foundation	Open lakehouse	1	3	Approved BigLake/Iceberg use cases and interoperability tests.	Data Platform
Data foundation	Reference/master data integration	1	4	Authoritative sources, survivorship interface and product contracts.	Data Governance
Engineering	SQL ELT	2	5	Dataform repositories, assertions, CI and release	Analytics Engineering

Domain	Capability	Current maturity	Target maturity	Target-state evidence	Accountable owner
				standards.	
Engineering	Streaming processing	1	4	Dataflow templates, load tests, state standards and runbooks.	Data Engineering
Engineering	Spark processing	2	4	Serverless job templates, dependencies, cost and operational standards.	Data Engineering
Engineering	Orchestration	2	4	Decision tree, DAG standards, monitoring and backfill procedures.	Data Platform
Governance	Enterprise catalog and glossary	1	5	Automated asset inventory, product metadata and governed definitions.	Data Governance
Governance	Lineage	1	4	Critical source-to-consumer lineage and change impact.	Data Governance
Governance	Data quality	2	5	Rule framework, scorecards, certification and issue workflow.	Domain Stewards
Governance	Data contracts	1	4	Versioned schemas, compatibility gates and consumer notifications.	Data Product Owners
Governance	Data marketplace and sharing	1	4	Certified listings, subscriber controls and usage telemetry.	Data Governance
Analytics	Semantic metrics	2	5	Approved KPI layer, tests, ownership and consumption patterns.	Analytics COE
Analytics	Self-service BI	2	4	Persona-based access, certified content and adoption support.	Analytics COE
AI	Managed MLOps	1	4	Pipelines, registry, deployment, monitoring and governance.	AI Platform
AI	Enterprise RAG and LLMOps	1	4	Document lineage, retrieval evaluation, safety, citations and cost.	AI Platform
Security	Data classification and DLP	2	5	Automated discovery, policy tags and classified-product controls.	Security/Privacy
Security	Key and secret management	2	5	CMEK policy, rotation, deletion protection and secret lifecycle.	Security Platform
Operations	Product SLOs and observability	1	5	Standard SLIs, dashboards, alerts, runbooks and error budgets.	SRE
Operations	Incident and problem management	2	4	On-call, incident command, post-incident review and trend actions.	SRE
Operations	Disaster recovery	1	4	Tiered recovery designs, scheduled tests and evidence.	BCDR/SRE
Economics	FinOps and allocation	1	5	Product/domain showback, forecasts, reservations and unit metrics.	FinOps
Delivery	Infrastructure as Code	2	5	Golden modules, controlled state, policy tests and drift detection.	DevSecOps
Delivery	CI/CD and software	2	5	WIF, immutable artifacts,	DevSecOps

Domain	Capability	Current maturity	Target maturity	Target-state evidence	Accountable owner
	supply chain			scans, approvals, provenance and rollback.	
Organization	Domain product ownership	1	4	Named owners, contracts, SLOs, roadmap and consumer review.	Domain Executives
Organization	Skills and community	1	4	Role-based learning, communities of practice and internal ownership.	Transformation Office

11.1 Maturity Scale

Level	Name	Characteristics
0	Absent	No recognized capability, owner or repeatable outcome.
1	Initial	Ad hoc delivery, individual knowledge and inconsistent controls.
2	Repeatable	Documented pattern used by multiple teams but with manual steps or variable evidence.
3	Defined	Enterprise standard, named owner, training, templates and measurable adoption.
4	Managed	Automated controls, SLOs, conformance and continuous improvement using telemetry.
5	Optimizing	Self-service, predictive operations, policy automation and demonstrated business unit economics.

11.2 Capability Dependencies

Outcome capability	Critical dependencies	Failure if dependency is weak
Certified data products	Domain ownership, contracts, catalog, quality, IAM and SLOs.	Products exist technically but are not trusted, supported or reusable.
Real-time decisions	Event contracts, Pub/Sub, Dataflow, source timestamps, monitoring and consumer actions.	Fast pipelines deliver ambiguous or unreliable data.
Self-service analytics	Curated products, semantic layer, identity and education.	Users revert to uncontrolled extracts and duplicate metrics.
Enterprise RAG	Governed documents, classification, chunk lineage, embeddings, retrieval evaluation and model controls.	Responses lack traceability or expose restricted information.
Regional resilience	Tier classification, region design, replication/replay, IaC and tested runbooks.	Recovery plan cannot meet business objectives.
Chargeback	Complete labels, product ownership, shared-cost policy and trusted billing data.	Financial disputes and counterproductive behavior.
Domain scale-out	Project factory, golden templates, delegated governance and internal skills.	Central platform becomes the throughput constraint.

11.3 Quarterly Capability Review

- Score maturity using objective evidence, not project-completion status.
- Identify capability constraints that block multiple domains and fund them as platform priorities.
- Retire capabilities and tools whose strategic purpose has ended.

- Track adoption, reliability, risk, lead time and unit economics together.
- Escalate capability ownership gaps to the executive steering committee.

12. Architecture and Transformation Roadmap

The roadmap sequences foundation, platform, domain adoption, migration, AI enablement and optimization. Timing is indicative and must be re-baselined after discovery, but the dependency order should remain unless risk is explicitly accepted.

Section outcomes

Phased delivery; Entry and exit criteria; Measurable outcomes; Decision gates.



Figure 12-1. Multi-wave architecture roadmap from mobilization through scaled domain adoption and optimization.

Phase	Timing	Objectives	Primary deliverables	Entry criteria	Exit criteria
Phase 0 — Mobilize and baseline	0–2 months	Approve charter, executive sponsorship, platform product ownership, assumptions, requirements, strategic ADRs and prioritized pilot domain.	Funding, decision rights, baseline architecture, source inventory, risk register and delivery backlog.	Sponsor and product owner named; critical decisions resolved; pilot has business owner and sources.	Mobilization complete; implementation teams can begin without unresolved identity, region or funding ambiguity.
Phase 1 — Secure cloud foundation	2–5 months	Deploy organization/folder/project patterns, Shared VPC, hybrid connectivity, identity federation, KMS, central logging, CI/CD bootstrap and policy guardrails.	Project factory, network products, IAM model, security baseline, evidence repository and Terraform state platform.	Approved landing-zone design; connectivity and identity dependencies available.	Security assurance passed; development and production projects provision through code.
Phase 2 — Minimum viable data platform	4–8 months	Implement landing/raw/quarantine/curated patterns, BigQuery standards, batch/CDC/streaming references, Dataform, orchestration, catalog, quality, monitoring and showback.	Golden modules, pipeline templates, metadata schema, SLO catalog, dashboards and first product environment.	Foundation stable; first-domain source and consumer contracts agreed.	Reference patterns pass performance, security, recovery and operational tests.
Phase 3 — First production domain	6–10 months	Deliver three to five certified products for a bounded domain, migrate priority consumers, operate SLOs and complete a recovery exercise.	Production products, Looker semantic model, runbooks, support model, product review and measured business outcome.	Product owner, stewards and engineers allocated; acceptance datasets and users available.	Three months stable operation; target adoption, quality and freshness achieved.
Phase 4 — Scale and federate	9–24 months	Onboard additional domains through standardized waves, delegate governance, expand shared capabilities, introduce reservations and mature FinOps.	Domain factories, marketplace, governance conformance, expanded pipeline/catalog coverage and cost allocation.	Pilot operating model accepted; platform backlog and support capacity funded.	Repeatable onboarding lead time; majority of new products use strategic patterns.
Phase 5 — Legacy modernization and retirement	12–36 months	Migrate prioritized warehouse, Hadoop, ETL and BI portfolios; reconcile, cut over and decommission hardware, software and contracts.	Migration factory, converted workloads, coexistence controls, retirement evidence and cost avoidance.	Target patterns proven; dependency inventory and retirement funding approved.	Legacy services retired according to portfolio plan; residual exceptions have funded exit paths.
Phase 6 — AI, automation and continuous optimization	12 months onward	Scale governed ML/RAG/agents, automate policy and evidence, improve predictive operations, optimize cost/performance and evolve products.	AI reference products, policy automation, higher maturity, product unit economics and continuous assurance.	Governed products and operational controls mature enough to support higher-risk use cases.	Quarterly value and maturity improvements; risk and unit cost remain within appetite.

12.1 Roadmap Work Packages

ID	Work package	Scope	Primary phases	Owner
WP-01	Executive governance	Charter, funding, steering cadence, architecture board and product council.	Phase 0	Executive Sponsor
WP-02	Organization and project factory	Folders, projects, billing, labels, policies and delegated administration.	Phases 1–2	Cloud Platform
WP-03	Network and hybrid connectivity	Shared VPC, IPAM, DNS, Interconnect/VPN, PSC, firewall and egress.	Phase 1	Network Platform
WP-04	Identity and security foundation	Federation, WIF, JIT, KMS, secrets, SCC, logs and perimeters.	Phases 1–2	Security Platform
WP-05	Data platform core	Storage zones, BigQuery, BigLake, service identities and governance integration.	Phase 2	Data Platform
WP-06	Pipeline factory	Batch, CDC, stream, SQL, Spark and orchestration templates.	Phases 2–3	Data Engineering
WP-07	Governance and product lifecycle	Catalog, glossary, lineage, contracts, quality, certification and marketplace.	Phases 2–4	Data Governance
WP-08	Analytics	Looker semantic layer, KPI governance, self-service and migration patterns.	Phases 3–5	Analytics COE
WP-09	AI platform	Vertex AI foundations, MLOps, RAG, evaluation and responsible-AI controls.	Phases 3–6	AI Platform
WP-10	SRE and BCDR	SLOs, dashboards, incident, capacity, recovery designs and exercises.	Phases 2–6	SRE/BCDR
WP-11	DevSecOps	Terraform modules, CI/CD, policy, provenance, tests and promotion.	Phases 1–6	DevSecOps
WP-12	FinOps	Allocation, budgets, forecasts, reservations, unit economics and optimization.	Phases 1–6	FinOps
WP-13	Domain adoption	Product ownership, sources, products, consumers, operations and value realization.	Phases 3–6	Domain Executives
WP-14	Migration factory	Assessment, conversion, reconciliation, cutover and decommissioning.	Phases 4–5	Migration Lead
WP-15	Change and skills	Role design, training, communities of practice and communications.	Phases 0–6	Transformation Office

12.2 Architecture Gates

Gate	Decision	Minimum evidence	Decision authority
G0 — Mobilization	Proceed with funded blueprint and pilot.	Charter, assumptions, requirements, risk, team and pilot business case.	Executive Steering Committee
G1 — Foundation design	Approve implementation of landing-zone baseline.	Organization/network/identity/security designs, threat model and ADRs.	Architecture and Security Review Boards

Gate	Decision	Minimum evidence	Decision authority
G2 — Platform readiness	Permit product development on shared platform.	Reference templates, monitoring, data controls, tests and support model.	Platform Product Council
G3 — Production readiness	Permit first product go-live.	Acceptance evidence, SLOs, runbooks, DR, cost forecast and business sign-off.	Product Owner, SRE, Security and Architecture
G4 — Scale decision	Increase domain onboarding concurrency.	Pilot stability, adoption, lead time, support load and unit economics.	Executive Steering Committee
G5 — Legacy cutover	Retire legacy service or wave.	Reconciliation, consumer sign-off, rollback, retention and decommission plan.	Business Owner and CIO delegate
G6 — AI risk approval	Deploy AI use case according to risk tier.	Evaluation, safety, privacy, model card, monitoring and human-oversight plan.	AI Governance and Risk

12.3 Roadmap KPIs

KPI	Baseline assumption	12-month target	24-month target	Owner
Governed project provisioning lead time	4–8 weeks	≤10 working days	≤2 working days	Cloud Platform
Standard source onboarding lead time	8–16 weeks	≤20 working days	≤10 working days	Data Platform
Certified products	Fragmented/unknown	10–20	50+	CDO and Domains
Golden-path adoption	Low	≥70% of new workloads	≥90%	Platform Product
Tier 0/1 SLO coverage	Inconsistent	100% of pilot	100% enterprise portfolio	SRE
Critical-product lineage	Partial	100% pilot	≥95% portfolio	Governance
Cost allocation	<60% assumed	≥90%	≥95%	FinOps
Legacy cost retired	No common baseline	Wave-specific committed savings	Portfolio target achieved	Migration Lead
Recovery exercises passed	Inconsistent	100% pilot Tier 0/1	100% Tier 0/1 annually	BCDR
Internal ownership	High external dependency assumed	Named internal owner per capability	Sustainable internal operating teams	Transformation Office

12.4 Critical Path and Dependencies

- Identity federation, organization hierarchy, billing and network connectivity precede production data-service deployment.
- Data classification and region strategy precede storage, replication and service-perimeter design.
- Project factory and CI/CD bootstrap precede scalable domain onboarding.
- Product ownership and source/consumer contracts precede pipeline implementation.
- SLO instrumentation, runbooks, cost allocation and recovery evidence precede production go-live.
- A stable first-domain operating model precedes high-concurrency migration waves.
- Governed data products and AI risk controls precede production generative-AI use of enterprise data.

12.5 Decision Calendar

Cadence	Forum	Decisions
Weekly during build	Platform architecture forum	Detailed design, standards, module changes and delivery blockers.
Biweekly	Data product council	Product prioritization, ownership, contracts and consumer readiness.
Monthly	Security and risk forum	Control evidence, exceptions, incidents and

Cadence	Forum	Decisions
		emerging service risks.
Monthly	FinOps review	Forecast, anomalies, reservations, optimization and allocation disputes.
Quarterly	Executive steering committee	Funding, value realization, maturity, major risks and roadmap changes.
Quarterly	Architecture Review Board	Strategic service lifecycle, ADR reviews, exceptions and technical debt.
Semiannual	Recovery and resilience review	Exercise results, dependency gaps and recovery investment.
Annual	Blueprint refresh	Framework, service, regulation and enterprise-strategy changes.

13. Enterprise Operating Model

The operating model assigns durable accountability for platform capabilities and domain data products. It combines a central platform product organization, federated domain product teams and independent assurance functions.

Section outcomes

Decision rights; Team topology; RACI; Support and lifecycle.

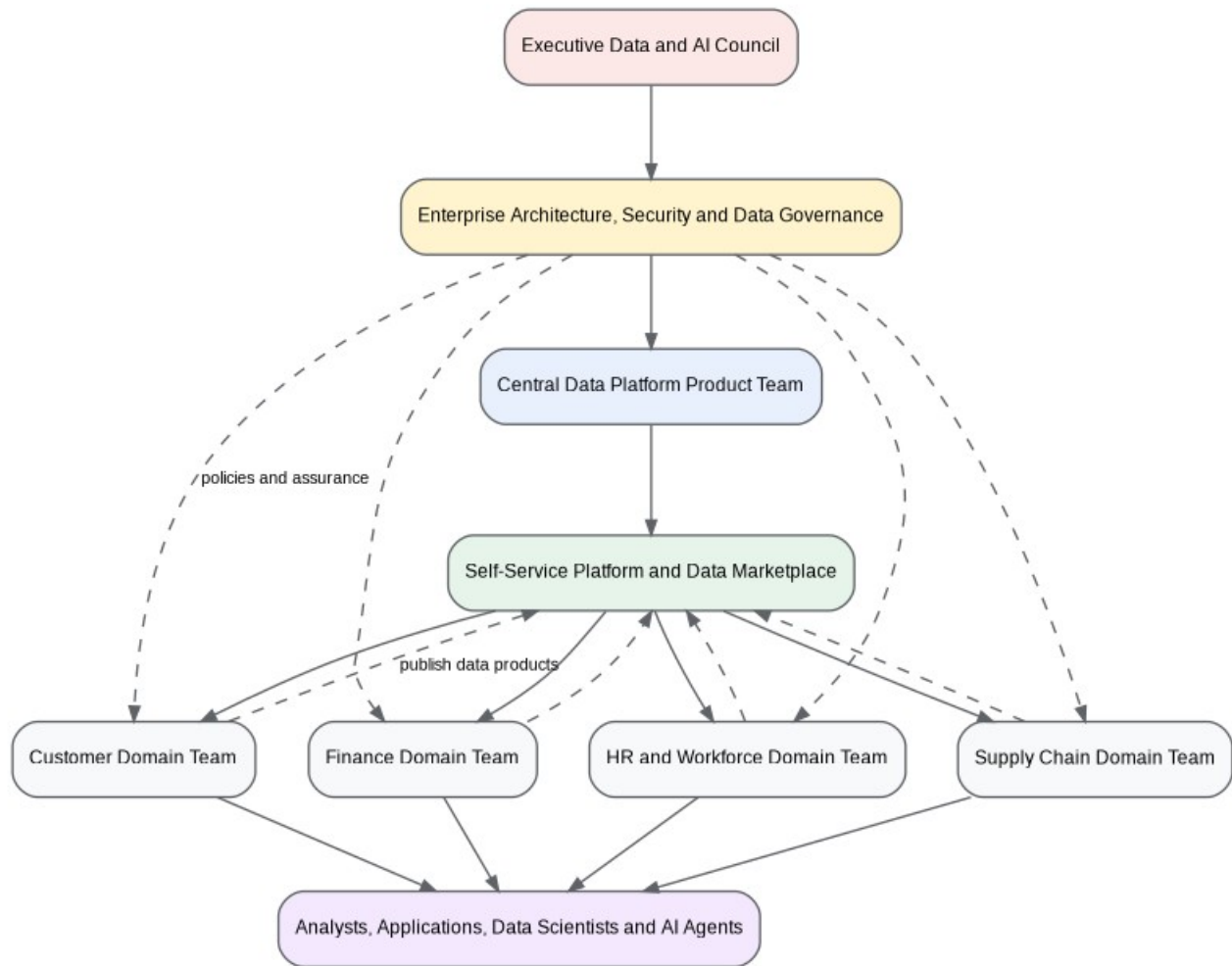


Figure 13-1. Federated operating model linking executive governance, central platform capabilities, domain product teams and assurance functions.

13.1 Team Topology

Team	Mission	Core responsibilities	Primary interfaces
Executive Steering Committee	Own enterprise outcomes, investment and cross-business decisions.	Funding, priority, risk appetite, ownership escalation and value realization.	CDO, CIO, CFO, CISO and business executives.
Platform Product Team	Deliver secure self-service data-platform capabilities.	Roadmap, product experience, service catalog, adoption, reliability and economics.	Domain teams, cloud foundation, SRE, governance and security.
Cloud Foundation Team	Operate organization, network, identity and shared services.	Project factory, Shared VPC, connectivity, policies, logging, KMS	Platform product, security, network

Team	Mission	Core responsibilities	Primary interfaces
		and platform IAM.	and DevSecOps.
Data Governance Office	Define policy and assure data trust.	Glossary, classification, stewardship, contracts, certification and conformance.	Domain owners, privacy, security and platform.
Domain Data Product Team	Own and operate business data products.	Source agreements, modeling, quality, SLO, support, consumer adoption and product roadmap.	Business owner, platform team and consumers.
Analytics COE	Provide governed semantic and analytical standards.	KPI governance, Looker standards, performance, content lifecycle and enablement.	Domains, executives and BI users.
AI Platform and Governance	Provide governed AI/ML capabilities and use-case assurance.	Vertex AI platform, MLOps/LLMOps, evaluations, responsible AI, model lifecycle.	Data products, risk, security and AI teams.
SRE and Operations	Protect production reliability and recovery.	SLOs, incidents, capacity, operational readiness, runbooks and recovery tests.	All service/product owners.
DevSecOps	Automate infrastructure and software delivery.	Terraform modules, CI/CD, policy, provenance, artifact and environment promotion.	Foundation, platform and domains.
FinOps	Make cost visible and optimize unit economics.	Allocation, forecasting, reservations, anomaly management and optimization.	Finance, platform and product owners.
Architecture and Assurance	Maintain standards and independent design review.	Reference architecture, ADRs, exceptions, conformance and lifecycle.	All delivery teams and review boards.

13.2 Decision Rights

Decision	Accountable	Responsible	Consulted	Escalation
Platform roadmap and funding	Executive Steering Committee	Platform Product Owner	Domains, Security, SRE, FinOps	CIO/CDO
Strategic service default	Architecture Review Board	Chief Architect	Platform, Security, Operations, FinOps	CTO
Domain/product ownership	Business-unit executive	Domain Product Owner	CDO, Governance	Executive Steering Committee
Data definition and quality target	Domain Data Owner	Data Steward/Product Team	Consumers, Governance	CDO
Access to restricted product	Domain Data Owner and Security	Access workflow	Privacy, Consumer Owner	CISO/CDO
Production go-live	Product Owner	Delivery Team	SRE, Security, Architecture, Governance	Steering Committee for residual high risk
SLO and error budget	Product Owner	SRE and Product Team	Consumers, Platform	Business Executive
CMEK and perimeter policy	CISO	Security Platform	Architecture, Network, Product	Security Review Board
Cost allocation and chargeback	CFO	FinOps	CDO, Domains, Platform	Executive Steering Committee
Architecture exception	Architecture Review Board	Requesting Owner	Security, SRE, FinOps as relevant	CTO/CISO for high risk

13.3 RACI for the Data-product Lifecycle

Activity	Business Owner	Product Owner	Data Steward	Platform	Security	SRE	Consumer
Prioritize product	A	R	C	C	I	I	C
Define contract and SLO	A	R	R	C	C	C	C
Classify data	A	R	R	I	C	I	I
Implement pipeline/product	I	A/R	C	C	C	C	I
Approve access	A	R	C	I	C	I	I
Certify product	A	R	R	C	C	C	I
Operate and remediate	I	A/R	R	C	C	R	I
Manage consumer change	A	R	C	I	I	I	C
Review cost	A	R	I	C	I	C	I
Deprecate product	A	R	C	C	C	C	C

13.4 Support Model

Support layer	Scope	Hours	Responsibilities	Handoff condition
L0 — Automation	Automated retries, self-healing, policy and runbook actions.	Continuous	Recover known transient failures and gather diagnostics.	Action fails or exceeds time/attempt threshold.
L1 — Operations	Monitoring, triage and standard procedures.	24x7 for Tier 0/1; business hours for lower tiers	Acknowledge, classify, execute runbook and communicate.	Product or platform expertise required.
L2 — Product/Platform Engineering	Pipeline, data, service and configuration diagnosis.	On-call by tier	Restore service, correct data, manage dependencies and implement safe fix.	Vendor or architectural defect suspected.
L3 — Specialist/Vendor	Google Cloud support, security, network, database or product specialist.	Contractual	Advanced diagnostics, product defect or critical design support.	Resolution or workaround available.
Problem Management	Recurring incident and structural risk elimination.	Business cadence	Root cause, trends, backlog and risk acceptance.	Corrective action closed and verified.

13.5 Product Lifecycle

27. Discover: identify consumer problem, owner, source authority, classification, expected value and service tier.
28. Design: define contract, interface, quality, SLO, region, security, cost model, failure modes and lifecycle.
29. Build: use approved modules and templates, create automated tests and register metadata.
30. Assure: complete architecture, security, privacy, governance, performance, cost and operational reviews.
31. Launch: migrate consumers, publish support information and begin SLO, adoption and cost measurement.
32. Operate: review quality, freshness, incidents, consumers, changes, cost and technical debt.
33. Evolve: version interfaces, communicate changes and preserve compatibility or provide migration.
34. Deprecate: notify consumers, restrict new use, archive required evidence and remove resources after approval.

13.6 Architecture Exception Lifecycle

Stage	Required action	Evidence
Request	State unmet requirement, selected alternative, owner and duration.	Exception request linked to design and ADR.
Assess	Evaluate security, reliability, interoperability, cost	Risk assessment and compensating controls.

Stage	Required action	Evidence
	and operational impact.	
Decide	Approve, conditionally approve or reject at the correct authority.	Recorded decision and expiry.
Implement	Apply controls, monitoring and isolation.	Test evidence and asset inventory.
Review	Reassess before expiry or after triggering change.	Updated risk and remediation status.
Close	Return to standard, adopt alternative as strategic, or retire capability.	Closure evidence and catalog update.

14. Enterprise Architecture Risk Register

The risk register focuses on risks introduced or materially affected by the reference architecture and transformation. Program risks are reviewed monthly; high and critical risks are escalated to executive governance.

Section outcomes

Risk ownership; Early indicators; Mitigation; Residual-risk decisions.

ID	Category	Risk	Probability	Impact	Rating	Early indicator	Mitigation	Owner
R-001	Ownership	Domains do not fund or perform product ownership, quality and support.	High	High	Critical	Unowned assets, unresolved quality issues, central backlog growth.	Executive accountability, product funding, RACI and certification gate.	CDO/Business Executives
R-002	Cost	Cloud consumption grows faster than business value or legacy cost retirement.	Medium	High	High	Budget variance, scan growth, dual-run duration and idle resources.	Showback, budgets, reservations, lifecycle, unit economics and retirement gates.	FinOps/CFO
R-003	Security	Restricted data is exposed through misconfigured access, export, AI or service integration.	Medium	Critical	Critical	Unexpected principals, egress, sensitive-data findings and unmanaged extracts.	Classification, least privilege, VPC SC, DLP, audit, AI controls and incident response.	CISO
R-004	Residency	Data is processed or replicated outside an approved geography.	Low/Medium	Critical	High	Cross-region transfer, resource-location drift or unapproved SaaS integration.	Residency policy, regional projects, transfer approvals and continuous scans.	Privacy/Legal
R-005	Reliability	Managed-service dependencies and recovery assumptions do not meet product targets.	Medium	High	High	Missed SLO, untested runbook, quota saturation or region dependency.	Tiered design, dependency objectives, load tests and recovery exercises.	SRE/BCDR
R-006	Migration	Legacy complexity, hidden consumers and reconciliation gaps delay retirement.	High	High	Critical	Wave slippage, rising exception count, repeated reconciliation failures.	Assessment factory, lineage/usage discovery, parallel validation and funded decommissioning.	Migration Lead
R-007	Data quality	Source defects and unclear semantics prevent product certification.	High	High	Critical	Low scorecards, steward backlog and consumer disputes.	Contracts, source-owner accountability, quarantine, quality gates and issue workflow.	Domain Owners
R-008	Platform bottleneck	Central teams cannot meet domain demand and workarounds proliferate.	Medium	High	High	Provisioning queue, shadow projects and declining template adoption.	Self-service, product SLOs, capacity plan, delegated governance and simpler golden paths.	Platform Product Owner
R-009	Technology sprawl	Exceptions and acquisitions recreate overlapping platforms.	Medium	High	High	Growing strategic-service catalog and aged exceptions.	Technology lifecycle, architecture governance, integration standards and retirement plan.	Chief Architect
R-010	Skills	Internal teams lack cloud, data-product, security and SRE expertise.	High	Medium/High	High	Consultant dependence, defects, slow incident resolution and low reuse.	Role-based enablement, paired delivery, communities and internal ownership targets.	Transformation Lead
R-011	Vendor dependency	Deep use of managed capabilities creates	Medium	Medium	Medium	Proprietary interfaces without export or exit plan.	Open formats where valuable, versioned	Chief Architect

ID	Category	Risk	Probability	Impact	Rating	Early indicator	Mitigation	Owner
		switching or negotiation risk.					code/data, export tests and ADR reviews.	
R-012	AI governance	Generative AI provides unsafe, inaccurate or unauthorized output.	Medium	Critical	Critical	Failed evaluations, missing citations, prompt injection or sensitive output.	Risk tiers, grounding, evaluations, guardrails, human oversight and audit.	AI Governance
R-013	laC	Terraform state or pipeline error creates destructive change at scale.	Low/Medium	Critical	High	Unexpected plan, drift, provider regression or state-access anomaly.	Protected state, plan/apply separation, tests, deletion protection and rollback.	DevSecOps
R-014	Observability	Teams monitor components but cannot detect product freshness or quality impact.	Medium	High	High	Healthy services with missed consumer commitments.	Product SLIs, end-to-end timestamps, quality metrics and dependency dashboards.	SRE/Product Owners
R-015	Compliance evidence	Controls exist but evidence cannot be reproduced for audit.	Medium	High	High	Manual screenshots, inconsistent logs or missing approvals.	Automated evidence, protected logs, control owner and sampled assurance.	Compliance
R-016	Change adoption	Business users retain legacy reports and extracts despite new products.	High	Medium	High	Low product adoption and continued duplicate metrics.	Consumer migration plan, KPI governance, training and legacy deprecation.	Business Owners
R-017	Quotas and scale	Service quotas or hot partitions constrain peak workloads.	Medium	High	High	Backlog, throttling, query queue or hot-key indicators.	Quota inventory, load tests, partition design, scaling requests and capacity review.	Platform/SRE
R-018	Data lifecycle	Retention, legal hold or privacy deletion is incomplete across derived copies.	Medium	Critical	Critical	Orphan copies, missed deletion evidence or conflicting policies.	Product lineage, policy metadata, controlled copies and end-to-end lifecycle tests.	Privacy/Governance

14.1 Risk Scoring and Escalation

Rating	Treatment	Governance
Critical	Immediate mitigation plan; no go-live without explicit executive risk acceptance.	Weekly review and executive escalation.
High	Funded mitigation with owner and target date; production conditions may apply.	Monthly architecture/program review.
Medium	Monitor indicators and implement proportionate controls.	Quarterly review.
Low	Accept or manage through standard controls.	Annual or change-triggered review.

14.2 Residual-risk Acceptance

- The risk owner must be senior enough to own the business impact, not only the technical implementation.
- Acceptance states scope, duration, financial or regulatory exposure, compensating controls and review trigger.
- Critical security, privacy and safety risks require the designated executive and control-owner approval.
- An accepted risk remains visible in dashboards and does not disappear from the register.
- Expired acceptance blocks further production change until renewed or remediated.

15. Architecture Assurance and Production Readiness

Architecture assurance validates that implementation preserves the reference architecture's intent. Reviews are evidence-based and proportional to data classification, service tier, scale and AI risk.

Section outcomes

Review lifecycle; Conformance criteria; Production readiness; Quality gates.

15.1 Assurance Lifecycle

Review point	Purpose	Required artifacts	Outcome
Concept review	Confirm business value, ownership, classification, tier and strategic fit.	Problem statement, product concept, consumer, source and initial risk.	Proceed, revise or stop before detailed design.
Architecture review	Validate logical/physical design, service choices, region, security, failure and cost.	Architecture diagrams, ADRs, data flows, threat model, NFR design and forecast.	Approved, conditional or exception required.
Detailed design review	Validate implementation-ready specifications and interfaces.	Schemas/contracts, IAM, network flows, Terraform plan, pipeline design and tests.	Ready to build or action list.
Security/privacy review	Validate control implementation and regulatory obligations.	Data classification, access model, perimeter, keys, logs, DLP and privacy workflow.	Security approval or remediation.
Performance/capacity review	Validate expected scale and quotas.	Load model, benchmark, query/partition design, autoscaling and quota plan.	Capacity approved.
Operational readiness review	Validate ownership and support.	SLOs, dashboards, alerts, runbooks, escalation, cost and dependencies.	Operations approval.
Recovery review	Validate RTO/RPO and service-specific recovery.	Recovery design, backup/replay, IaC, test evidence and failback.	Recovery approval.
Production readiness review	Confirm all evidence and residual risk.	Acceptance matrix, release/rollback, business sign-off and open conditions.	Go, conditional go or no-go.
Post-launch review	Validate actual outcome, stability and cost.	SLO, incidents, quality, adoption, cost and user feedback.	Stabilize, optimize or remediate.

15.2 Mandatory Production-readiness Checklist

Area	Production-readiness requirement	Pass evidence
Ownership	Business owner, product owner, technical owner, data steward and on-call owner are named.	Evidence link or test result recorded in the product assurance record.
Requirements	Functional and non-functional requirements map to implemented controls and tests.	Evidence link or test result recorded in the product assurance record.
Classification	Data classification, residency and service tier are approved and propagated.	Evidence link or test result recorded in the product assurance record.
Architecture	Conceptual, logical, physical, data-flow and trust-boundary diagrams are current.	Evidence link or test result recorded in the product assurance record.
ADRs	Material decisions and deviations are recorded and approved.	Evidence link or test result recorded in the product assurance record.
Identity	Human and workload identities are least privilege; no prohibited long-lived credentials exist.	Evidence link or test result recorded in the product assurance record.
Network	Ingress, egress, DNS, private access, perimeter and firewall behavior are tested.	Evidence link or test result recorded in the product assurance record.
Encryption	At-rest and in-transit encryption are confirmed;	Evidence link or test result recorded in the product

Area	Production-readiness requirement	Pass evidence
	CMEK lifecycle is tested where required.	assurance record.
Data lifecycle	Retention, archive, legal hold and privacy deletion are implemented and tested.	Evidence link or test result recorded in the product assurance record.
Quality	Quality rules, thresholds, issue workflow and publishing gates are active.	Evidence link or test result recorded in the product assurance record.
Contracts	Schemas/interfaces, compatibility, versioning and deprecation are documented.	Evidence link or test result recorded in the product assurance record.
Lineage	Critical source-to-product and product-to-consumer lineage is visible.	Evidence link or test result recorded in the product assurance record.
Performance	Representative volume, burst and concurrency tests meet targets.	Evidence link or test result recorded in the product assurance record.
Quotas	Relevant service quotas and scaling limits are documented and monitored.	Evidence link or test result recorded in the product assurance record.
SLOs	Availability, freshness, quality and latency SLIs/SLOs are measured.	Evidence link or test result recorded in the product assurance record.
Observability	Dashboards, actionable alerts, ownership and runbook links are complete.	Evidence link or test result recorded in the product assurance record.
Incident	Escalation, incident command and communication procedures are tested.	Evidence link or test result recorded in the product assurance record.
Recovery	RTO/RPO design and recovery/failback tests meet the product tier.	Evidence link or test result recorded in the product assurance record.
CI/CD	Build, tests, provenance, approvals, deployment and rollback are demonstrated.	Evidence link or test result recorded in the product assurance record.
Security testing	Threat scenarios, negative access tests and vulnerability findings are resolved.	Evidence link or test result recorded in the product assurance record.
Compliance evidence	Control mappings, approvals and logs are reproducible.	Evidence link or test result recorded in the product assurance record.
Cost	Forecast, budget, allocation metadata and optimization controls are active.	Evidence link or test result recorded in the product assurance record.
Consumer readiness	Access, documentation, support, training and migration are complete.	Evidence link or test result recorded in the product assurance record.
Legacy impact	Coexistence, rollback and decommissioning obligations are understood.	Evidence link or test result recorded in the product assurance record.
Residual risk	Open conditions have owners, dates and approved risk acceptance.	Evidence link or test result recorded in the product assurance record.

15.3 Conformance Scorecard

Domain	Weight	Conformant criteria
Business and product	10%	Outcome, ownership, consumer, roadmap and lifecycle are approved.
Data and governance	15%	Layers, contracts, quality, metadata, lineage and certification comply.
Security and privacy	20%	Identity, classification, network, encryption, audit and privacy controls pass.
Reliability and recovery	15%	SLO, capacity, failure modes, RTO/RPO and tests pass.
Engineering and delivery	15%	IaC, CI/CD, tests, versioning and rollback comply.
Operations	10%	Monitoring, alerts, runbooks, incident and support are ready.

Domain	Weight	Conformant criteria
FinOps	5%	Forecast, allocation, budget and unit metrics are active.
Architecture and interoperability	10%	Strategic services, interfaces, ADRs and exceptions are valid.

A Tier 0 or Tier 1 product must be conformant in every critical security, privacy, reliability and recovery item. An aggregate score cannot offset failure of a mandatory control. Conditional approval records evidence due dates and may restrict scale or consumer access.

15.4 Reference Architecture Maintenance

- Review this volume at least annually and after material Google Cloud service, regulatory or enterprise-strategy change.
- Track official service renames and retire deprecated terminology from diagrams, code and standards.
- Validate regional availability, quotas, SLA, CMEK and VPC Service Controls support for each implementation.
- Promote repeated approved exceptions into the strategic catalog only after enterprise evaluation.
- Retire obsolete patterns and publish migration guidance rather than retaining multiple permanent golden paths.
- Maintain alignment across detailed landing-zone, data platform, security, AI, DevSecOps, FinOps, SRE and Terraform volumes.

16. Executive Decisions and Prioritized Actions

The blueprint can proceed only when decision rights, funding, ownership and the initial transformation scope are explicit. The following actions convert the reference architecture into an executable enterprise program.

Section outcomes

Executive decisions; First 180-day actions; Expected evidence; Next-volume dependencies.

16.1 Executive Decisions Required

ID	Decision	Authority	Required output
ED-01	Approve the central platform product and federated domain ownership model.	CIO and CDO	Platform charter, RACI and funded owners.
ED-02	Approve regional residency zones and the initial primary/recovery region pairs.	CISO, Privacy and Regional CIOs	Residency matrix and legal assessment.
ED-03	Approve BigQuery, Cloud Storage/BigLake, Dataflow, Pub/Sub, Dataform, Knowledge Catalog, Looker and Vertex AI as strategic defaults.	CTO and Architecture Board	Strategic service catalog and exception policy.
ED-04	Approve classification-based CMEK and VPC Service Controls for restricted-data zones.	CISO	Security control standard.
ED-05	Select and fund the first production domain and three to five products.	Business Executive and CDO	Product charters, source/consumer commitments and benefit baseline.
ED-06	Approve Tier 0–3 SLO/RTO/RPO model and residual-risk authority.	CIO and Business Continuity	Service-tier policy.
ED-07	Approve platform funding, showback and chargeback-readiness approach.	CFO and CIO	Funding model and FinOps roadmap.
ED-08	Approve legacy modernization and decommissioning as funded scope, not optional follow-up.	CIO and CFO	Migration waves and benefit ownership.
ED-09	Approve internal capability-building and named long-term owners.	CHRO/CIO	Role and enablement plan.
ED-10	Approve AI risk-tier governance before production enterprise-data use.	Chief AI Officer, CISO and Risk	AI policy and review process.

16.2 First 180 Days

Timing	Workstream	Action	Output
Days 0–30	Mobilize	Confirm sponsor, product owner, architecture board, assumptions, pilot domain and delivery teams.	Approved charter and decision calendar.
Days 0–45	Discovery	Inventory sources, data classes, users, products, pipelines, reports, costs and dependencies.	Prioritized assessment backlog and baseline metrics.
Days 15–60	Foundation design	Complete organization, project, network, identity, KMS, logging, perimeter and region designs.	Approved detailed design and Terraform backlog.
Days 30–90	Bootstrap delivery	Establish source repositories, WIF, build runners, artifact storage, state and policy tests.	Auditable development pipeline.

Timing	Workstream	Action	Output
Days 45–120	Deploy minimum foundation	Provision development/test foundations, hybrid connectivity, central logs and security controls.	Foundation acceptance evidence.
Days 60–150	Build reference data patterns	Implement batch, CDC, streaming, SQL and quality templates with failure tests.	Validated reference implementations.
Days 60–150	Governance foundation	Define glossary, classifications, product manifest, contracts, certification and access workflow.	Governance operating procedures and catalog baseline.
Days 90–180	Pilot product build	Deliver first products, semantic model, consumer access, SLOs, dashboards, runbooks and cost forecast.	Production-readiness evidence or bounded remediation plan.
Days 120–180	Recovery and security exercises	Run perimeter, negative-access, incident and recovery tests.	Exercise reports and closed critical findings.
Days 150–180	Scale decision	Evaluate product outcome, platform lead time, risk, support and economics.	Decision to scale, adjust or pause.

16.3 Expected Deliverables from Subsequent Volumes

Volume	Detailed deliverable
Volume 02 — Landing Zone	Organization, projects, Shared VPC, hybrid network, identity, security foundation, logging and implementation examples.
Volume 03 — Enterprise Data Platform	Ingestion, storage, BigQuery/BigLake, metadata, quality, contracts, products and detailed data flows.
Volume 04 — Data Engineering	Dataflow/Beam, Spark, Dataform, orchestration, reusable pipelines, standards and code.
Volume 05 — Data Mesh	Domain model, product lifecycle, federated governance, marketplace and example domains.
Volume 06 — AI Platform	Vertex AI, MLOps, LLMOps, RAG, agents, evaluations, responsible AI and reference code.
Volume 07 — Security Blueprint	Zero Trust, IAM, network/data perimeters, KMS, DLP, threat models and compliance control mappings.
Volume 08 — DevSecOps Automation	Repositories, CI/CD, tests, provenance, promotion, rollback and policy-as-code.
Volume 09 — FinOps	Billing, allocation, reservations, budgets, unit economics and optimization.
Volume 10 — Operations SRE	SLIs/SLOs, observability, incidents, capacity, runbooks and recovery operations.
Volume 11 — Terraform Reference	Golden modules for foundation, platform and domain resources.
Volume 12 — Industry Implementations	Healthcare, banking, retail and telecom reference implementations.
Volume 13 — Architecture Playbook	Review templates, decision trees, checklists, workshops and adoption guidance.

Recommended immediate scope

Fund and execute the landing-zone detailed design, minimum viable data platform, and first production domain as one coordinated tranche. Separating these into unrelated projects creates ownership gaps and delays evidence of business value.

Appendix A. Enterprise Assumptions

The reference architecture uses the following planning assumptions because organization-specific discovery has not yet occurred. They are not presented as confirmed facts. Each must be validated, rejected or refined during mobilization.

ID	Area	Assumption	Architecture consequence
A-001	Enterprise scale	Global Fortune 100 organization with more than 100,000 employees and multiple legal entities.	Drives federated ownership, regional zones and enterprise governance.
A-002	Data history	10–50 PB of historical structured, semi-structured and unstructured data.	Requires migration waves, tiered storage and portfolio prioritization.
A-003	Daily growth	50–250 TB of new data per day plus billions of events.	Requires elastic services, quota planning and scale tests.
A-004	Source estate	Hundreds of databases, SaaS applications, mainframes, APIs, files, Kafka and cross-cloud sources.	Requires multiple governed ingestion patterns.
A-005	Consumers	Thousands of analysts, hundreds of scientists and operational applications.	Requires workload isolation, semantic governance and access automation.
A-006	Geographies	Operations across the Americas, Europe, Asia-Pacific and other regulated regions.	Requires residency classification and regional service validation.
A-007	Cloud posture	Hybrid and multi-cloud coexistence for several years.	Requires private connectivity, controlled transfer and authoritative-source management.
A-008	Identity	Enterprise identity provider can federate to Google Cloud.	Enables group-based IAM, MFA and short-lived workload access.
A-009	Compliance	GDPR, HIPAA, PCI DSS, SOC 2, ISO 27001 and NIST-aligned controls may apply by workload.	Requires overlay control mapping rather than a single universal certification claim.
A-010	Service tiers	Tier 0–3 model is acceptable as the initial reliability taxonomy.	Drives SLO, support, encryption, logging and recovery investment.
A-011	Operating model	Central platform engineering and domain teams can receive explicit ownership and funding.	Required for data-mesh and product model.
A-012	Delivery	Terraform and source-controlled CI/CD are permitted strategic approaches.	Enables repeatable deployment and recovery.
A-013	Analytics strategy	BigQuery and Looker can be evaluated as strategic defaults.	Enables warehouse and semantic consolidation.
A-014	AI strategy	Vertex AI can be evaluated for governed ML and generative AI.	Enables common AI lifecycle and control patterns.
A-015	Commercial model	Current Google Cloud pricing, licensing and support terms will be assessed during detailed design.	Prevents use of invented or outdated price assumptions.
A-016	Network	At least two resilient hybrid connectivity paths can be established for critical regions.	Supports private migration and operations.
A-017	Data ownership	Business executives can nominate accountable data-product owners and stewards.	Required for quality, access and lifecycle decisions.
A-018	Legacy retirement	The program has authority and funding to decommission systems after migration.	Required for benefits realization.
A-019	Security operations	Enterprise SOC and incident processes can integrate Google Cloud findings and logs.	Required for operational security.
A-020	Documentation lifecycle	Architecture and product metadata	Required to keep the blueprint aligned

ID	Area	Assumption	Architecture consequence
		will be managed as living artifacts.	with implementation.

Appendix B. Global Naming and Metadata Conventions

B.1 Resource Naming Pattern

Names should be deterministic, concise and stable. Sensitive business meaning should not be embedded in resource names. Where Google Cloud resource length or character rules differ, the implementation module applies an approved abbreviation.

Resource	Pattern	Example	Notes
Project	prj-{domain}-{capability}-{env}-{region-code}	prj-cust-data-prod-in1	Project IDs remain globally unique; append controlled suffix if needed.
Dataset	{domain}_{layer}_{purpose}_{region}	customer_curated_profile_in	Use lowercase and underscores; security boundary may justify separate dataset.
Bucket	bkt-{org}-{domain}-{layer}-{env}-{region}-{suffix}	bkt-acme-customer-raw-prod-in-01	Avoid sensitive names; enable uniform access.
Pub/Sub topic	tpc-{domain}-{event}-{version}-{env}	tpc-order-created-v1-prod	Event contract version included.
Subscription	sub-{consumer}-{event}-{env}	sub-fraud-order-created-prod	Consumer-owned; define DLQ and retention.
Service account	sa-{workload}-{purpose}-{env}	sa-cust-ingest-prod	One purpose; avoid shared identities.
KMS key ring	kr-{classification}-{region}-{env}	kr-restricted-in-prod	Region and separation of duties aligned.
KMS key	key-{service}-{purpose}	key-bq-customer	Rotation and deletion protection policy.
Dataflow job	df-{domain}-{pipeline}-{env}	df-customer-cdc-prod	Template/release version recorded in labels.
Airflow environment	af-{platform-or-domain}-{env}-{region}	af-data-prod-in	Managed Service for Apache Airflow naming.
Data product ID	dp.{domain}.{product}.{major-version}	dp.customer.golden_profile.v1	Stable enterprise identifier independent of physical table.
Label/tag	{key}={controlled-value}	data_product=customer360	Mandatory keys enforced by modules/policy.

B.2 Mandatory Resource Metadata

Metadata key	Purpose	Example value	Control
business_domain	Cost and ownership grouping.	customer	Controlled vocabulary.
data_product	Product-level allocation and traceability.	golden_profile	Matches catalog identifier.
environment	Control and lifecycle.	prod	dev, int, test, stage, prod, dr, sandbox.
owner	Accountable team.	grp-customer-data	Resolves to managed group/team directory.
cost_center	Financial allocation.	CC10420	Validated against finance master.
classification	Data-control tier.	restricted	public, internal, confidential, restricted.
service_tier	Reliability and recovery tier.	tier1	tier0 to tier3.
residency	Approved geography.	in	Controlled region/geography code.

Metadata key	Purpose	Example value	Control
managed_by	Provisioning mechanism.	terraform	Exceptions reported.
expiry_date	Automatic lifecycle for temporary assets.	2027-03-31	Required for sandbox/temporary resources.
criticality	Operational importance.	business_critical	Used for alert and change policy.
compliance_scope	Applicable overlay.	pci	Multiple values represented through approved mechanism.

Appendix C. Global Security Classification

Class	Definition	Examples	Minimum controls
Public	Approved for unrestricted external disclosure.	Published reports, public product information.	Integrity, ownership and release approval.
Internal	Routine enterprise information not intended for public release.	Internal operational reference, procedures.	Enterprise identity, least privilege where appropriate, logging.
Confidential	Disclosure could harm customers, employees or business operations.	Customer records, financial performance, contracts.	Need-to-know access, audit, approved regions, encryption, non-production controls.
Restricted	Disclosure or loss could create severe legal, safety, regulatory or financial impact.	Payment data, health data, national IDs, highly sensitive HR/security information.	VPC SC where supported, selective CMEK, fine-grained access, DLP, strict egress, enhanced audit and recovery.

Classification is inherited by derived data unless an approved transformation demonstrably reduces sensitivity. Aggregation, masking or tokenization does not automatically change classification; the data owner and security/privacy functions approve reclassification.

Appendix D. Traceability Index

Outcome	Drivers	Requirements	Principles	Key ADRs	Roadmap
Business acceleration	BD-01, BD-02, BD-11	BR-ING, BR-DAT, BR-ANA	AP-01, AP-10, AP-15	ADR-001, ADR-002, ADR-010	Phases 2–4
Legacy modernization	BD-03, BD-12	BR-DAT, BR-OPS	AP-06, AP-07, AP-20	ADR-002, ADR-003, ADR-016	Phases 4–5
Real-time operations	BD-04	BR-ING	AP-01, AP-09, AP-12	ADR-008, ADR-009	Phases 2–4
Security and privacy	BD-05, BD-06	BR-SEC	AP-03, AP-04, AP-05	ADR-004 to ADR-007, ADR-017	Phases 1–6
Data products and mesh	BD-07	BR-DAT, BR-GOV	AP-02, AP-10, AP-18	ADR-001, ADR-012, ADR-013	Phases 2–4
AI readiness	BD-08	BR-AI	AP-03, AP-19	ADR-015, ADR-017	Phases 3–6
Reliability	BD-09	BR-OPS and NFR-AVL/DR	AP-12, AP-13	ADR-018, ADR-020	Phases 2–6
Cost and sustainability	BD-10	BR-OPS and NFR-CST	AP-06, AP-14, AP-16	ADR-019	Phases 1–6

Appendix E. Glossary

Term	Definition
ADR	Architecture Decision Record.
AI-ready	Data, security, operating and delivery capabilities that support governed AI—not merely access to a model API.

Term	Definition
Authoritative source	The approved system or product that owns a fact at a defined business boundary.
CMEK	Customer-managed encryption key controlled through Cloud KMS or supported key-management pattern.
Data contract	Versioned agreement covering schema, semantics, quality, SLO, classification and change behavior.
Data mesh	Operating and architecture approach based on domain ownership, data as product, self-service platform and federated governance.
Data product	A discoverable, owned, supported and governed data interface that serves defined consumers.
Error budget	The permitted unreliability implied by an SLO, used to balance change and stabilization.
Golden path	Approved reusable implementation pattern designed to be the easiest compliant route.
Knowledge Catalog	Google Cloud's metadata, discovery and governance capability; formerly Dataplex Universal Catalog.
Lakehouse	Architecture combining open object-storage data with warehouse-style management and analytics capabilities.
Managed Service for Apache Airflow	Google Cloud's managed Airflow orchestration service; formerly named Cloud Composer.
RAG	Retrieval-augmented generation, which grounds model output in retrieved enterprise or external content.
Residency zone	A governed geography and region pattern in which data may be stored, processed and recovered.
Service tier	Business-impact category that determines SLO, support, control and recovery requirements.
SLI	Service Level Indicator: a measured aspect of service or product behavior.
SLO	Service Level Objective: a target value or range for an SLI.
VPC SC	VPC Service Controls, a managed-service perimeter control used to reduce data-exfiltration risk for supported services.
WIF	Workforce or Workload Identity Federation, depending on context; used to access Google Cloud without long-lived keys.
Zero Trust	Access model based on explicit identity and context verification, least privilege and continuous evaluation.

Appendix F. Authoritative Reference Set

Detailed designs must validate current behavior against official documentation. The following reference set informed the blueprint baseline and should be reviewed during implementation.

Reference	Official location
Google Cloud Architecture Framework	https://cloud.google.com/architecture/framework
Google Cloud enterprise foundations blueprint	https://cloud.google.com/architecture/security-foundations
Google Cloud landing zone design	https://cloud.google.com/architecture/landing-zones
Google Cloud Adoption Framework	https://cloud.google.com/adoption-framework
BigQuery documentation	https://cloud.google.com/bigquery/docs
Cloud Storage documentation	https://cloud.google.com/storage/docs

Reference	Official location
BigLake documentation	https://cloud.google.com/bigquery/docs/biglake-intro
Pub/Sub documentation	https://cloud.google.com/pubsub/docs
Dataflow documentation	https://cloud.google.com/dataflow/docs
Datastream documentation	https://cloud.google.com/datastream/docs
Dataform documentation	https://cloud.google.com/dataform/docs
Managed Service for Apache Airflow documentation	https://cloud.google.com/composer/docs
Knowledge Catalog and Dataplex documentation	https://cloud.google.com/dataplex/docs
Vertex AI documentation	https://cloud.google.com/vertex-ai/docs
VPC Service Controls documentation	https://cloud.google.com/vpc-service-controls/docs
Cloud KMS documentation	https://cloud.google.com/kms/docs
Sensitive Data Protection documentation	https://cloud.google.com/sensitive-data-protection/docs
Cloud Monitoring and Logging documentation	https://cloud.google.com/monitoring/docs
FinOps Foundation Framework	https://www.finops.org/framework/

Product names, quotas, regional availability, previews, pricing, SLAs, CMEK integration, VPC Service Controls support and lifecycle status can change. Implementation teams must confirm the exact feature and region in current official documentation and record material findings in solution ADRs.

Appendix G. Document Completion Statement

Volume 01 completion

This volume establishes the executive architecture baseline, measurable requirements, non-functional targets, principles, strategic reference architecture, technology defaults, decision records, capability model, roadmap, operating model, risk register and assurance approach. Detailed implementation specifications are delegated to the subsequent blueprint volumes while remaining governed by the decisions in this document.